

BIOL 108: BIOLOGY FOR NON-MAJORS LAB MANUAL



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Detailed Licensing

Detailed Licensing

Licensing

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CHAPTER OVERVIEW

1: Lab Safety

[1.1: Introduction](#)

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1.1: Introduction

Goals

- Familiarize students with basic safety rules.
- Allow students to become familiar with the safety equipment in the laboratory.

Learning Objectives

Upon completion of this lab, students will be able to:

- Identify the location and proper use of emergency equipment.

Introduction

Laboratory safety involves all the measures taken by the laboratory worker, laboratory owner, institution, and regulatory agencies to eliminate potential harm to human health and well-being. Although steps are taken to reduce risks in the workplace, safety is a matter of personal responsibility. A biotechnology lab may have several safety hazards that must be known and understood by all students or employees working in the lab. It is the responsibility of each person in the lab to know and follow basic laboratory safety rules, understand how to safely operate equipment, understand the hazards of materials they are working with, and reduce potential risks. In the event of an accident, it is critical to know the location and use of emergency equipment. Having this knowledge should help to prevent accidents and minimize the damage that might occur in the event of an accident.

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1.2: Laboratory Safety Equipment

Draw the Table in your lab notebook. Fill in the location of the following lab safety items:

Table: Laboratory Safety Equipment and Location

Safety Equipment	Location
Telephone and Campus Emergency Number	
Fire Alarm	
Fire Extinguisher	
Eye Wash	
Glass Waste Container	
Disinfectant	
Goggles	
Gloves	
First Aid Kit	
Broom and Dust Pan	
Emergency Class Evacuation site	

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1.3: Laboratory Safety Guidelines

General Rules

- No guests are allowed in the lab, including children or minors.
- Know emergency procedures, use, and location of emergency equipment (emergency exits, fire extinguishers, fire blanket, eye wash station, first aid kit, and broken glass container).
- In case of fire, evacuate the room and assemble outside the building.
- Report all accidents, no matter how insignificant they appear, to a laboratory supervisor
- Be aware of your surroundings and potential dangers created by others.

Personal Protection

- Do not smoke, eat, drink, chew gum, or apply cosmetics in a laboratory.
- Wear protective clothing such as long pants, closed-toe shoes, a lab coat, and goggles.
- Tie long hair up or behind the shoulders. Do not wear long, dangling jewelry or scarves.
- Dispose of gloves in the laboratory trash. Do not wear lab coats and gloves in public areas. You will need to dispose of gloves and take off your lab coat before leaving the lab.
- Cover cuts or scrapes with a sterile, waterproof bandage before entering a lab.
- Wash skin immediately and thoroughly if contaminated by chemicals or microorganisms.
- Wash your hands regularly, with soap and water, especially after working with things that could be contaminated with bacteria.
- If you have immediate health concerns, please let your lab instructor know immediately.
- Let your lab instructor know before leaving the classroom.

Handling Chemicals

- Keep all containers capped with the appropriate lid. Clearly label items produced in the lab with contents, your name, and the date.
- If a chemical is splashed into the eyes or skin, flush for 15 minutes.
- Clean up spills and broken glass immediately. Use a broom and dustpan to pick up broken glass.
- Keep chemicals away from direct heat or sunlight. Keep containers of alcohol, acetone, and other flammable liquids away from flames.
- Read labels carefully. Be aware of hazardous chemicals and precautions for safe use.
- Follow instructions about the proper disposal of lab reagents.

Handling Equipment

- Keep your work area clean and clutter-free.
- Be aware of your potential impact on others.
- Notify the lab supervisor or instructor of malfunctioning equipment.
- If you do not know how to use an instrument or equipment, then do not touch it.
- Do not use laboratory equipment without first receiving instruction in its use.
- Keep balances clean and dry, always use weigh paper/boats.
- Never leave heat sources unattended. Be careful when using hot plates or burners. Note that there is often no visible flame, glow or sign that objects are hot.

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1.4: Lab Safety Contract

Copy this statement into your lab notebook and all group members should sign the page.

I have read the lab safety guidelines, found all lab safety equipment, and understand the procedure for emergency class evacuation. I will conduct myself in a safe and conscientious manner and take proper care in the use of all lab equipment.

Signature _____ **Date** _____

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CHAPTER OVERVIEW

2: Lab 1- Science and Policy

[2.1: Overview and Purpose](#)

[2.2: Part One- The Scientific Method](#)

[2.3: Part Two- Termite Experiment](#)

[2.4: Activity 2- Environmental Laws and Regulations](#)

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2.1: Overview and Purpose

The purpose of this laboratory exercise is to familiarize students with the scientific method and to demonstrate how to apply it to real-world problems. Through this exercise, students will gain an understanding of the components of the scientific method and how to use it to develop hypotheses, design experiments, and interpret data. Additionally, there may be a discussion about an environmental issue, environmental law, and policy stemming from these laws.

Learning Outcomes

- The student will be able to use steps of the scientific method and devise, carry out, and report an experiment using the scientific method
- The student will be able to clearly present and communicate data and results
- The student will be able to write a lab report for a simple experiment that follows the scientific method
- (optional) The student will be able to analyze an environmental policy from the perspective of different stakeholders

Learning Objectives

- Students will be able to explain the components of the scientific method.
- Students will be able to formulate hypotheses based on research.
- Students will be able to design experiments that test their hypotheses.
- Students will be able to analyze and interpret data from their experiments.
- Students will be able to draw conclusions from their experiments.

Vocab

Hypothesis

Experiment

Data

Analysis

Conclusion

Scientific Method

Variables

dependent variable

Independent variable

Control

Controlled variables

Pheromones

Policy

Environmental law

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2.2: Part One- The Scientific Method

All rigorous scientific inquiry follows the scientific method. Understanding this process is essential when devising and carrying out your own experiments and critically analyzing the results of other experiments. There are two types of scientific inquiry: discovery-based and hypothesis-based. Discovery-based science is often exploratory or descriptive, whereas hypothesis-based inquiry is used to answer ‘how’ and ‘why’ questions. Hypothesis-based science follows the steps scientific method.

The scientific method is a systematic process that scientists use to gather information and test their ideas about the world. It typically involves the following steps:

1. Observing and asking questions: Scientists observe the natural world and ask questions about what they observe with their senses. Both types of scientific inquiry use this.
2. Forming a hypothesis: A hypothesis is a testable explanation for the observations, it is NOT an educated guess. Scientists form a hypothesis based on their observations and background knowledge.
3. Designing an experiment: Scientists design an experiment to test their hypothesis. They must control all other variables so that they can isolate the effects of the variable they are studying.
4. Collecting data: Scientists collect data by performing their experiments. They measure and record the results. Being able to measure something is a key component of science. Scientists have invented and developed many tools to aid in making measurements with a high degree of accuracy and measuring things that are impossible to measure with just human senses.
5. Analyzing data: Scientists analyze their data to see if it supports or disproves their hypothesis. They look for patterns and relationships in their data.
6. Drawing conclusions: Based on the data, scientists draw conclusions about the validity of their hypothesis. If the data support the hypothesis, scientists must form a new hypothesis and start the process over again.
7. Communicating results: Scientists communicate their results to others through publications, conferences, and other means. Other scientists can then attempt to replicate the results to verify their validity.

To write a good hypothesis, first answer your question: is it ‘yes’, or ‘no’? Then, explain why you think this is the answer. Then write this into a clear statement. For example, you may suspect that the abundance of birds is related to the number of trees. You think this is because more trees = more food and nesting sites:

‘Bird abundance increases with the number of trees in a habitat because more trees increase the amount of food and nesting sites.’

Note

You must explain HOW bird abundance varies with tree density. A mediocre hypothesis would be ‘bird density changes with tree density’ because it doesn’t address how it may happen and doesn’t account for increases or decreases. Another example that could use improvement might be: ‘the number of leaves in a forest positively correlates to bird abundance’ because it doesn’t try to explain why that happens not to mention that it would be extremely difficult to make an experiment to test this.

Test your understanding: hypothesis

You notice that your dog prefers chicken to vegetables. Research tells you that dogs were selectively bred from wolves: efficient predators with a meat-based diet.

Write a hypothesis for this observation:

Make a **prediction**. There are many ways to test your hypothesis. You could count the number of birds you see in an hour; you could count the number of nests or feathers you find; put out bird feeders and count how much food is eaten per bird visit... likewise, there are many ways of counting the number of trees.

Your prediction is what you think you will see in the ACTUAL RESULT(S) that you will collect in YOUR experiment if your hypothesis is not proven wrong. It also helps to restate the hypothesis in your prediction:

Example predictions:

If your methods were to walk in a forest and observe the number of birds seen in a day and then count all of the trees, your prediction would be: ‘if bird density increases with tree density then more birds will be counted in one day in habitats with more trees compared to habitats with fewer trees.’

If instead you put out bird feeders and used the amount of food eaten to indicate how many birds there are, your prediction would be: ‘if bird density increases with tree density then more food will be eaten from bird feeders in habitats with more trees compared to habitats with fewer trees.’

Test your understanding: Prediction

You test whether dogs prefer chicken to vegetables by giving dogs two bowls of food: one with 200g of chicken, the other with 200g of boiled vegetables. After 1 minute you record how much food is left in each bowl.

Write a prediction for this experiment based on the hypothesis you wrote earlier:

Carry out an **experiment**. Use accurate and precise methods that enable you to collect unbiased, accurate data and test your hypothesis. A **variable** is anything that can be measured or controlled in an experiment. **An independent variable is a variable that you will manipulate. The dependent variable is the variable being tested and that often ‘depends’ i.e. responds to the independent variable.** In our example, we think that the number of birds depends on the number of trees. The **independent** variable is the **number of trees** and the **dependent** variable is the **number of birds**.

A **control variable** is something that is kept constant, such as temperature, or the type of binoculars you use to observe birds. A **control group** is a group in which you will compare your experimental group with. For example, if you wanted to test the impact of eating sugar on tooth decay, your control group will be the group not given extra sugar, whereas the experimental group is given extra sugar.

Only test **one** variable at a time! Otherwise, you will not be able to see what is impacting your dependent variable. If we tested both tree density and insect abundance at the same time, we would not know which one was affecting bird density.

Test your understanding: variables

Identify the independent and dependent variables for your experiment testing food preferences in dogs:

Independent:

Dependent:

7. Form a **conclusion**. What did you find? What does your data show? What is the answer to your question? Your conclusion is always based on your results and analyses and states whether your hypothesis and predictions were supported. **Remember: it is ok to not support your hypothesis or to not reach a concrete conclusion!** This is all part of the scientific process. If your hypothesis is disproven, go back and formulate and test a new one! This is often the time when you would write your entire experiment into a lab report.

6. **Analyze** your data. You now need to examine and analyze your findings to find out what they mean. This could be as simple as plotting a graph of your results or calculating means and ranges. The method you chose should reveal the answer to your question. For our example, we could plot a graph of the number of trees vs. the number of birds in an area.

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2.3: Part Two- Termite Experiment

Learning Objectives

- Students will observe the behavior of termites and draw conclusions about their behavior.
- Students will record data as to termite behavior.
- Students will analyze data and discuss a summary of their results.
- Students will recognize elements in the experimental design to include a hypothesis, variable to be tested, control, and other variables.

Background Information

Termites are small, soft-bodied, usually pale-colored insects. These are social insects with a caste system. They live in colonies on the ground or in wood. Their food consists primarily of wood or other vegetable material. The workers are reproductively sterile and lack compound eyes. They do the main work of the colony-collecting food, feeding the queen, soldiers, and young, and constructing galleries.

Many species cause considerable damage to buildings, furniture, utility posts, fence posts, and other materials. In addition, termites are important decomposers breaking down dead trees and other plant materials into nutrients useful to plants. The cellulose in the termite's food is digested by protists living in the termite's digestive tract. This association is an excellent example of symbiosis or mutualism. Termites frequently groom each other with their mouthparts, a result of the attraction of secretions that are usually available on the body.

Problem: Observe the behavior of termites and make conclusions about their behavior. **Work together in groups of 3-5:**

Materials: Petri dishes, Paper, various ballpoint pens, colored pencils, Termites, small soft-bristled paint brushes, and scissors.

Initial Procedure:

1. Cut several pieces of paper to fit the inside of your Petri dishes by tracing the Petri dish and cutting out the circle so that it fits in the bottom of the dish.
2. Draw a large figure eight on one of the papers with one of the ballpoint pens provided (go over it several times with the pen) and place the paper in the petri dish. Repeat this on a different piece of paper with a colored pencil.
3. Using a paintbrush, CAREFULLY brush 3-4 termites onto the paper in each dish and observe their behavior.
4. Develop a chart or table to record your data in a way that measures what you are seeing (These will be quantitative data). Example: (termite response) +, (no response) 0, (termite avoids the markings) -. There are other ways that you may want to record what you are seeing. Be sure to include a column in the chart to record any observations (This is qualitative data).
5. Discuss the termites' behavior with your group. Did you notice anything unusual that may be useful for further study? After a general discussion, each group of students will discuss an experimental question involving the termites' behavior. Once your group has developed a question, discuss it with the instructor before proceeding.
6. Now, suggest a hypothesis to be tested in an experiment. Write your hypothesis below.
7. Develop an experiment based on your question. Test as many other pens and pencils as needed to test your hypothesis and record your data in the table that you have made. Be very thoughtful and intentional about this step. Ask yourself: 'will using this pen test for more than one variable?', 'Am I changing things in the experiment that will test for things not related to my hypothesis?', etc.
8. Carry out your experiment and record your results. Draw conclusions about your observations and give a scientific explanation as to why the termites respond as they do.
9. Repeat these steps with a revised hypothesis until you think you know what is going on.

DISCUSSION/QUESTIONS:

1. Were any errors or other unforeseen factors that affected the experiment(s) present?
2. How would you modify possible future experiments?
3. Why should we conclude that the hypothesis can be either rejected or supported, rather than proven? Can you prove a hypothesis to be completely correct?

4. Why does this experiment work? How might you test this?
5. What purpose do termites serve in the natural world?

Now consider this:

Termites use pheromones for communication and to coordinate their behavior within their colony. Pheromones are chemical signals that termites release to communicate with each other. There are several types of pheromones that termites produce for different purposes, including

1. Trail pheromones: Termites use trail pheromones to lay down a chemical trail from the colony to a food source. Other termites follow the trail and feed on the food source.
2. Alarm pheromones: When a termite encounters danger, it releases an alarm pheromone to warn other termites in the colony. The alarm pheromone triggers a defensive response in the other termites.
3. Aggregation pheromones: Aggregation pheromones are used by termites to gather in groups. They are produced by young termites and attract workers to their location.
4. Queen pheromones: Queen pheromones are produced by the queen termite and attract workers to her location. They also suppress the development of other reproductive individuals in the colony.

By using pheromones, termites can coordinate their behavior and ensure the survival of the colony.

1. Does this information change your thinking about your experiment?
2. If you read this entire lab before starting the experiment, congratulations! You researched preexisting knowledge before you began experimenting with living organisms. This is a great example of why education and extensive research is so vitally important. It can change how you think about a new problem that you've not encountered and draw rational conclusions with far less guesswork.

If you read the information about pheromones before class, did it help you formulate a hypothesis?

If you did not read the pheromone information before lab, do you think it may have been helpful?

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2.4: Activity 2- Environmental Laws and Regulations

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Introduction

In general, Policy is a set of principles, rules, and guidelines formulated or adopted by an organization to reach its long-term goals and to guide its decision-making. Laws and policies are related in that policies are created to support the laws that have been put in place. Policies provide the framework for how laws will be implemented and enforced, as well as what is expected of those who are governed by the law. Although environmental laws are generally considered a 20th-century phenomenon, attempts have been made to legislate environmental controls throughout history. In 2,700 B.C., the Middle Eastern civilization in Ur passed laws protecting the few remaining forests in the region. In 80 A.D., the Roman Senate passed a law to protect water stored for dry periods so it could be used for street and sewer cleaning. During American colonial times, Benjamin Franklin argued for "public rights" laws to protect the citizens of Philadelphia against industrial pollution produced by animal hide tanners.

Significant environmental action began at the beginning of the 20th century. In 1906, Congress passed the "Antiquities Act," which authorizes the president to protect areas of federal lands as national monuments. A few years later, Alice Hamilton pushed for government regulations concerning toxic industrial chemicals. She fought, unsuccessfully, to ban the use of lead in gasoline. She also supported the legal actions taken by women who were dying of cancer from their exposure to the radium then used in glow-in-the-dark watch dials. During the early 1960s, biologist Rachel Carson pointed out the need to regulate pesticides such as DDT to protect the health of wildlife and humans.

With the establishment of the Environmental Protection Agency (EPA) in 1970, environmental law became a field substantial enough to occupy lawyers on a full-time basis. Since then, federal and state governments have passed numerous laws and created a vast network of complicated rules and regulations regarding environmental issues. Moreover, international organizations and agencies including the United Nations, the World Bank, and the World Trade Organization have also contributed to environmental rules and regulations. Because of the legal and technical complexities of the subjects covered by environmental laws, persons dealing with such laws must be knowledgeable in the areas of law, science, and public policy. Environmental laws today encompass a wide range of subjects such as air and water quality, hazardous wastes, and biodiversity. The purpose of these environmental laws is to prevent, minimize, remedy, and punish actions that threaten or damage the environment and those that live in it. However, some people believe that these laws unreasonably limit the freedom of people, organizations, corporations, and government agencies by placing controls on their actions.

Possible activities:

California State senate

For this exercise, we will act out a potential debate at a meeting based on a real issue in the western United States. This debate will be based on "Counting Sheep" by Elizabeth Clark (instructor handout) and/or the document produced by The Wild Sheep Foundation. For this activity, you will split into senatorial groups based on different roles, research the issue, and formulate a position on the issue based on your assigned role.

Your group's role:

Your group's position:

Newscast

In small groups, learn the different aspects of the debate and try and write a script outlining as many sides to the story as possible. There should be about 5 short paragraphs produced, one about the history, one about the sheep, one about the lions, and two about management controversies.

When complete, hand your paragraphs to your instructor who will act as a managing editor. Your instructor may put the paragraphs from different groups up for vote and discussion. You can choose the best paragraphs and sentences produced by all the groups to make the best possible newscast covering the topic.

Questions for after the activity:

1. Why do we preserve a particular species? Because of its value to the conservation of overall biodiversity, or because of an intrinsic right to exist?
2. Why is it socially or ethically acceptable to hunt certain species but not others? Where and why do we draw the line?
3. What are the pros and cons of culling lions? What are some methods that the Fish and Game Commission could use to moderate the cons?
4. What are the cons of bighorn sheep extinction?
5. Humans hunt wild ungulates and rodents. Why do voters have such a strong aversion to killing lions when hunting other mammals is morally or socially acceptable? Is this concern for charismatic megafauna actually counterproductive to environmental goals/biodiversity conservation? Briefly explain your answer.

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CHAPTER OVERVIEW

3: Lab 2 - Human Genetics

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[3.2: Phenotype Analysis](#)

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3.1: Overview and Purpose

Overview and Purpose

Students will observe single gene traits and relate the observations to possible genotypes of individuals. Inheritance patterns are discussed and students will analyze pedigrees to identify inheritance patterns of traits within a population. Human karyotypes will be observed and chromosomal differences will be correlated with human chromosomal syndromes. Students will gain a basic understanding of how phenotypes are determined by alleles on chromosomes and the differences in inheritance patterns of traits.

Learning Outcomes

1. Identify possible genotypes of individuals from observed phenotypes
2. Visually identify abnormalities in human karyotypes
3. Identify genotypes of individuals within human pedigrees



Learning Objectives

- Observe human single gene traits within a group
- Correlate phenotypes of single gene traits to possible genotypes
- Analyze pedigrees to determine patterns of inheritance
- Analyze karyotypes to identify chromosomal abnormalities

Vocabulary

Chromosome

Homologous chromosomes

Gene

Allele

Dominant allele

Recessive allele

Homozygous

Heterozygous

Genotype

Phenotype

Autosomes

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3.2: Phenotype Analysis

Background Information

Most cells making up the human body contain a nucleus. Inside of the nucleus are chromosomes that are responsible for transmitting hereditary information from cell to cell in cell division and from parent to offspring in reproduction. Chromosomes are thread-like structures that consist of tightly coiled DNA (deoxyribonucleic acid), the genetic material, and associated proteins. DNA essentially serves as the blueprints for life. DNA is organized into genes. Genes (i.e., "packets of DNA") control the structure and function of all cells, as well as large-scale processes such as development and reproduction. Most genes encode for a specific protein. Proteins play a variety of different essential roles in the body, including acting as enzymes, allowing communication between cells, immune functions, regulation of molecules into and out of cells, contraction of muscles and many more.

Humans have 23 pairs of chromosomes (46 total chromosomes) in the nucleus of all somatic cells (non-sex cells). However, the sex cells (sperm cells and eggs) contain only one copy of each chromosome. Human reproduction requires a sperm cell to fuse with an egg during fertilization to create the zygote. The zygote contains 23 pairs of chromosomes. Twenty-two pairs of the chromosomes are autosomes, containing genes that are unrelated to sex. The 23rd pair of chromosomes are the sex chromosomes, X and Y. Females have two X chromosomes and males have an X and a Y chromosome. A single gene on the Y chromosome, SRY (sex-determining region Y), triggers an embryo to develop male physical characteristics.

Chromosomes that are the same length and contain the same genes are called homologous chromosomes. Humans have 22 pairs of homologous chromosomes, receiving one of each pair from the mother and one of each pair from the father. Each chromosome contains thousands of individual genes. There are thought to be about 20,000 genes in human DNA. Interestingly, the number of genes in human DNA is not appreciably different from the number of genes in chimpanzees or mice.

Some human traits are determined by a single gene, or *mostly* determined by a single gene. Different versions of a gene allow for variations in a population. The different versions of a gene are called alleles. For example, earlobe shape is controlled by a single gene. One allele for earlobe shape causes individuals to have free earlobes and another allele causes people to have attached earlobes. Humans have two alleles for each gene (receiving one allele from our mom and one allele from our dad). If both alleles are the same version then this is referred to as homozygous. When an individual has two different alleles for the same gene this is referred to as heterozygous. The alleles that an individual has are referred to as the genotype. But, the outward expression of the alleles (what we see) is the phenotype. The phenotype that is expressed in heterozygous individuals is the dominant phenotype and is caused by the dominant allele. The recessive allele is the phenotype that is not seen in the phenotype of heterozygous individuals. The dominant allele is generally represented by a capital letter, while the recessive allele is generally represented by a lowercase letter. The alleles for earlobe shape are E and e. The dominant allele, E, determines free earlobes. Homozygous dominant (EE) and heterozygous (Ee) individuals have the free earlobe phenotype. But, homozygous recessive (ee) individuals have the recessive attached earlobe phenotype.

Most traits are determined by more than one gene. For example, skin color and height are determined by many genes. A few human phenotypes however, are determined by a single gene. We will explore some of these single gene traits in the laboratory.

Materials needed

1. A group of students to observe phenotypes

Exercise 1: Phenotype Analysis

1. Use a lab partner to help you determine your phenotype for the traits listed.
2. Complete the table below. Use two alleles per trait for the genotype. If you exhibit the dominant phenotype, use a dash to represent the second allele.

Example:

B_ genotype for the phenotype of brown eyes (dash indicates second allele could be B or b which means a genotype of BB or Bb)

bb genotype for the phenotype of blue eyes

Interlocking fingers

Interlock fingers. Observe which thumb is on top (right or left). The tendency to place the left thumb on top is due to a dominant allele (I) and the genotype is I- (either II or I-). The right thumb on top is determined by the ii genotype.

Ear lobes

The dominant allele (E) results in the phenotype of free earlobes. The recessive allele (e) is for attached earlobes.

Widow's peak

Widow's peak occurs when the hairline forms a distinct point in the center of the forehead. Widow's peak is controlled by a dominant allele (W).

Tongue curling

A dominant allele (T) gives the individual the ability to curl the tongue in a U-shape.

Hitch hiker's thumb

A person homozygous recessive for this trait (hh) can bend the last (distal) thumb joint back to about a 90 degree angle. Those with the H allele cannot.

PTC tasting

If you can taste the bitterness of PTC, you have the dominant allele (P). Place the PTC paper on your tongue for a few seconds. If you cannot taste anything, you do not possess the dominant allele.

Straight hair

A person with the dominant allele (H) has straight hair. Homozygous recessive individuals have curly hair.

Handedness

If you are right-handed then you have the dominant allele (R). The recessive allele (r) encodes left-handedness.

Mid-digital hair

A person with the dominant allele (H) has hair on the middle joint of any finger. People with no hair on the middle finger joint are homozygous recessive.

Number of students in class _____

Trait and alleles	My phenotype is: (dominant or nondominant)	My genotype is (X_ or xx)	# in class with dominant phenotype	% of class with dominant phenotype
Interlocking fingers (I or i)				
Earlobes (E or e)				
Widow's peak (W or w)				
Tongue curling (T or t)				
Hitchhiker's thumb (H or h)				
PTC tasting (P or p)				
Straight hair (H or h)				
Handedness (R or r)				
Mid digital hair (h or h)				

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3.3: Karyotype Analysis

Background Information

After the sperm fertilizes the egg the zygote begins dividing to generate more cells. Errors during replication of the chromosomes can result in breakages, deletions or duplications in the number of chromosomes in each cell. Sometimes the chromosomes in the egg or sperm are already altered even before fertilization occurs. Typically, an embryo missing large portions of chromosomes or essential genes will not result in a live birth. However, there are many chromosomal differences, such as extra chromosomes, deletions of chromosomes or deletions of parts of chromosomes, that do not have lethal effects. The phenotypic effects of the chromosomal differences may be significant or subtle, depending on the genes that are added, altered or missing. Many of these chromosomal alterations can be identified by examining chromosomes in the form of a karyotype. A human karyotype is an arrangement of the autosome pairs from longest to shortest, followed by the sex chromosomes as the last arranged pair.

Below are descriptions of some human chromosomal syndromes.

Down syndrome, occurring in 1 in 900 live births, is the most common chromosomal syndrome in humans. This condition is also called Trisomy 21 because it results in three copies (instead of the normal 2 copies) of chromosome 21. Individuals with Down syndrome often have physical growth delays, intellectual disability and congenital heart defects.

Edwards syndrome, also known as Trisomy 18, results from a third copy of all or part of chromosome 18. This affects about 1 in 2,500 pregnancies, contributes to stillbirths, resulting in the syndrome prevalence of 1 in 6,000 live births. Babies born with Edwards syndrome are often born smaller in size, have heart defects, clenched fists with overlapping fingers, smaller heads and jaws, deformed feet, and severe intellectual disabilities.

Turner syndrome is a condition occurring in 1 in every 2,000-4,000 female live births, in which a female is missing some or all of an X chromosome (X0). Individuals with Turner syndrome often have short stature, extra folds of skin on the neck, and many are born with heart defects. Affected girls with Turner syndrome do not undergo expected sexual changes during puberty, including development of breasts and ovaries, without the assistance of hormonal treatment. Individuals with Turner syndrome have normal intelligence. Depending on severity of symptoms, diagnosis may occur before birth, during infancy or childhood, or during the teen and adult years.

Fragile X syndrome is characterized by changes in a gene FMR1, on the X chromosome. Most people have few (5-40) repeats of a CGG sequence in this gene. But, people with Fragile X syndrome have more than 200 repeats within the gene. This causes mild to moderate intellectual disability and behavioral challenges.

Klinefelter syndrome occurs in males who have an extra X chromosome, resulting in XXY genetic makeup. Klinefelter syndrome occurs in 1 in 500-1,000 male births. Affected individuals have male sex organs, but poor sexual development during puberty. Affected males often produce lower testosterone, have reduced body muscle mass and body hair, have enlarged breast tissue and are often sterile.

Prader-Willi syndrome results when genes on chromosome 15 are not functional. In most cases this is due to the deletion of part of chromosome 15 received from the father. Most common symptoms of Prader-Willi syndrome are behavioral problems, intellectual disability and short stature. Due to hormonal imbalances, people with Prader-Willi syndrome are often constantly hungry, leading to obesity and diabetes. Prader-Willi syndrome affects about 1 in every 15,000 births.

Materials needed

Human karyotypes provided by your instructor.

Exercise 2: Karyotype Analysis

Method: Your instructor will provide your lab group with several karyotypes. Examine the karyotypes with your lab group. Answer the questions for each karyotype.

1. Karyotype 1

- Is the individual biologically male or female? _____
- What chromosomal abnormality is shown? _____

c. The likely syndrome is _____

2. Karyotype 2

a. Is the individual biologically male or female? _____

b. What chromosomal abnormality is shown? _____

c. The likely syndrome is _____

3. Karyotype 3

a. Is the individual biologically male or female? _____

b. What chromosomal abnormality is shown? _____

c. The likely syndrome is _____

4. Karyotype 4

a. Is the individual biologically male or female? _____

b. What chromosomal abnormality is shown? _____

c. The likely syndrome is _____

5. Karyotype 5

a. Is the individual biologically male or female? _____

b. What chromosomal abnormality is shown? _____

c. The likely syndrome is _____

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3.4: Pedigree Analysis

Background Information

Recall that autosomal chromosomes are all the chromosomes in a genome that are not the sex chromosomes. For disorders that are caused by a **dominant allele** there are some trends that you will see in a population of a family tree that carries that allele. Since you only need 1 copy of a dominant allele to express a dominant trait, autosomal dominant disorders often will occur in every generation of a family. If two individuals who do not have the disorder reproduce, it is impossible for any of their children to be affected by the disorder. Huntington's disease is an example of a disease caused by an autosomal dominant allele.

If the gene (and its resulting disorder) reduces the chance of surviving or reproducing, its frequency should decrease over time. This is because if the disorder results in death before a person reaches reproductive age, they will never get the chance to pass the deleterious allele to the next generation. This is not always the case however, because some disorders do not become evident until after the individual has reached reproductive age, and already unknowingly passed the disorder to their children. This is the case with Huntington's disease, a debilitating and deadly disease that leads to a serious degeneration of the nervous system, and does not appear until the age of 30 or later. In other instances, such as achondroplasia, this disorder does not affect the person's life expectancy significantly or impact normal reproductive events, and will therefore persist generation after generation.

Autosomal recessive inheritance means that a person can be a **carrier** of the deleterious allele, **but not be affected** with the disorder. Cystic fibrosis, phenylketonuria (PKU), and albinism are all examples of autosomal recessive traits. This inheritance pattern has characteristic trends in a population. Heterozygotes are typically symptom-free carriers and only homozygous recessive individuals are affected. Although there are exceptions to this rule, such as sickle cell anemia, where heterozygotes display some symptoms, but they are so mild that it generally does not affect their overall life expectancy. Sickle cell anemia is an example of a disorder that displays codominance. Because inheritance of an autosomal recessive disorder requires 2 copies of the allele to be expressed, these types of disorders tend to skip generations. Two heterozygous parents have a 50 percent chance of producing heterozygous children and a 25 percent chance of producing a homozygous recessive child. When both parents are homozygous recessive, then all children will be affected.

Sex-linked traits: Disorders that are carried on the X chromosome have very different patterns in a population. Male offspring will always inherit the only X chromosome they carry from their mother. Males cannot inherit the X chromosome from their father, since the paternal X chromosome would result in a female offspring. Clues that a trait is a recessive X-linked:

- More males than females are affected because males only get one shot at an X chromosome. They cannot be carriers for a disorder, they either have it, or they don't. Females need two copies, therefore their chances of inheriting both deleterious alleles is greatly diminished compared to males.
- An affected father cannot pass trait to sons, and in order for a female to be affected by a sex-linked disorder, her father **MUST** express the disorder as well.
- X-linked dominant disorders are often fatal, and do not result in viable offspring that survive outside of the womb and are exceedingly rare. For this reason, we will not address X-linked dominant disorders.
- A note on X-linked annotation. With autosomal alleles, the correct annotation is to use capital letters for dominant and lower case for recessive. Genes found on the X chromosome however, are annotated a bit differently. Capital and lowercase are still used, but most often these are added as a sub- or superscript next to the X. In this example Punnett square for example, the mother who is a carrier for, let's say color blindness, would be annotated XBX^b .

Exercise 3: Pedigree Analysis

Inheritance patterns of traits can often be determined by analyzing a **pedigree**, a chart that depicts several generations of parents and their children and illustrates the individuals that are affected by a particular trait. In a pedigree males are depicted with squares and females with circles. When an individual is affected by the trait being studied then their square or circle is black. When the trait is absent from an individual then their square or circle is white. Parents are connected by a horizontal line and children are shown beneath the parents, connected to the parents with a vertical line. Each generation is shown on the same horizontal line.

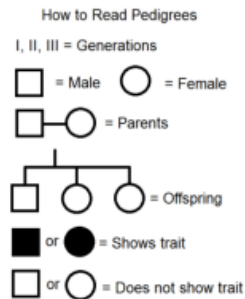
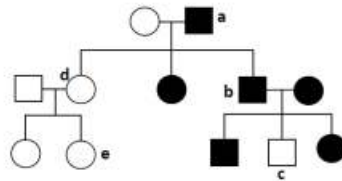


Figure X. How to read pedigrees. “BI-102 General Biology 2” by Linn Benton Virtual College. 7/2020, CC BY.
<https://www.oercommons.org/courses/b...biology-2/view>

Pedigree 1

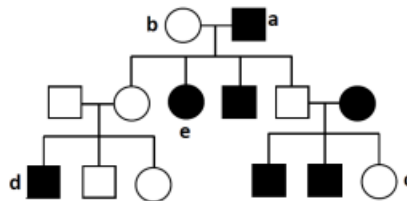


Things to note visually in this pedigree:

- Both male and females are affected at about the same percentage
- A large percentage of offspring at each generation are affected
- There are no offspring with the disease that don't have at least one affected parent

1. Is this disorder autosomal or sex-linked?
2. Is this disorder most likely caused by a dominant or recessive gene?
3. Circle the individual(s) that proves your choices.
4. What are the genotypes of the individuals labeled in the pedigree?
 - a. Grandfather:
 - b. Son:
 - c. Grandson 1:
 - d. Daughter 1:
 - e. Granddaughter 1:

Pedigree 2



Things to note visually in this pedigree:

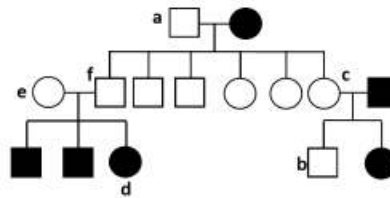
- A much higher percentage of males are affected by this disorder.
- Females are only affected if their father is also affected.
- An offspring is affected by the disorder even though both parents are not.

1. Is this disorder autosomal or sex-linked?
2. Is this disorder most likely caused by a dominant or recessive gene?
3. Circle the individual(s) that proves your choices.

4. What are the genotypes of the individuals labeled in the pedigree?

 - a. Grandfather:
 - b. Grandmother:
 - c. Granddaughter 1:
 - d. Grandson 1:
 - e. Daughter 1:

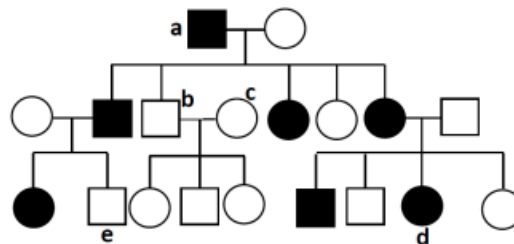
Pedigree 3



1. Is this disorder autosomal or sex-linked?
2. Is this disorder most likely caused by a dominant or recessive gene?
3. Circle the individual(s) that proves your choices.
4. What are the genotypes of the individuals labeled in the pedigree?

 - a. Grandfather:
 - b. Grandson 1:
 - c. Daughter 1:
 - d. Granddaughter 1:
 - e. Daughter-in-Law:
 - f. Son 1:

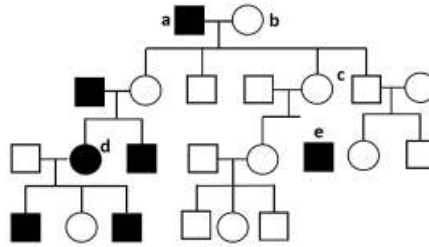
Pedigree 4



1. Is this disorder autosomal or sex-linked?
2. Is this disorder most likely caused by a dominant or recessive gene?
3. Circle the individual(s) that proves your choices.
4. What are the genotypes of the individuals labeled in the pedigree?

 - a. Grandfather:
 - b. Son 1:
 - c. Daughter-in-Law:
 - d. Granddaughter 1
 - e. Grandson 1

Pedigree 5



1. Is this disorder autosomal or sex-linked?
2. Is this disorder most likely caused by a dominant or recessive gene?
3. Circle the individual(s) that proves your choices.
4. What are the genotypes of the individuals labeled in the pedigree?
 - a. Grandfather:
 - b. Grandmother:
 - c. Daughter 1:
 - d. Granddaughter 1:
 - e. Grandson 1:

Questions for Review

1. In typical situations, how many chromosomes are inherited from our mother? How many from our father?
2. What does it mean if someone is a carrier for a trait?
3. Why is it that even though some dominant disorders are fatal, they continue to persist in human populations?
4. Explain why disorders caused by a dominant allele cannot have carriers of the disorder.
5. Why is it possible for two individuals who express a dominant disorder to have children who are unaffected?
6. A son receives his X-chromosome from:
7. Explain why men cannot be carriers for an X-linked trait.

Testing Understanding Post-Lab

1. Imagine you are a genetic counselor, and a couple who is planning on starting a family comes to you for information. Charles was married once before, and he and his first wife had a child who suffered from cystic fibrosis (an autosomal recessive disorder). His current wife Elaine's brother died of cystic fibrosis, but both parents did not have the disorder.
 - a. What is Charles' genotype?
 - b. What is the genotype of Elaine's parents?
 - c. What is the genotype of Elaine's brother?
 - d. What is the chance that Elaine is a carrier for the disorder?
2. A couple is both phenotypically normal, but their son suffers from hemophilia, an X-linked recessive disorder. What percent of their children are likely to suffer from hemophilia? What percent are likely to be carriers for the disorder?
3. Roger and Gayle are both of normal phenotype but have a son with Muscular Dystrophy, a sex-linked recessive disorder. Gayle is pregnant and amniocentesis shows the fetus is male. What is the probability that the fetus will suffer from muscular dystrophy?

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3.5: References

"Book: Unfolding the Mystery of Life - Biology Lab Manual for Non-Science Majors, "Lab 8 Exercies: Human Genetics and Cytogenetics," (Genovesi, Blinderman & Natale)" by Ellen

Genovesi, Laura Blinderman, & Patrick Natale, LibreTexts is licensed under CC BY .<https://bio.libretexts.org/@go/page/24114>

"BI-102 General Biology 2" by Linn Benton Virtual College. 7/2020, CC BY. <https://www.oercommons.org/courses/b...biology-2/view>

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CHAPTER OVERVIEW

4: Lab 3 - Biotechnology

4.1: Overview and Purpose

4.2: DNA Extraction

4.3: Analyze Ingredient Label for GMO Ingredients

4.4: Test Food Products for GMO Genes

4.5: Gene Therapy

4.6: References

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4.1: Overview and Purpose

Pre-lab homework- bring a food ingredient label or a picture of a food ingredient label to the lab.

Overview and Purpose

This lab is designed to provide a basic understanding of biotechnology concepts that students are likely to encounter and hear about in modern society. This includes biotechnological methods used in agriculture and medicine.

Learning Outcomes

1. Explain the process of a DNA extraction procedure
2. Define genetically modified organism
3. Describe how biotechnology is used in agriculture and medicine
4. Describe the process of CRISPR
5. Explain what knowledge is gained through microarray analysis

Learning Objectives

- Perform a DNA extraction procedure
- Analyze a food label for possible genetically modified ingredients
- Use lateral flow test strips to detect genetically modified ingredients in foods

Vocabulary

Recombinant DNA technology

Gene therapy

Genetically Modified Organism

CRISPR/Cas9

Microarrays

Hydrophilic

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4.2: DNA Extraction

Background Information

Within the chromosomes of organisms, specific sections of DNA, called genes, code for specific proteins. A gene is “read” through a process of transcription to generate a sequence of mRNA. The mRNA is then used as the blueprint instructions to generate a specific protein during a process called translation. The process of generating a protein from a gene is often referred to as gene expression (Figure 1). A change, or mutation, in DNA can generate a protein that has a different amino acid sequence and therefore an altered or malfunctioning protein. Mutations in DNA of human chromosomes that cause malfunctioning or absent proteins are often the cause of genetic diseases. This lab discusses biotechnological advances that are used to detect and treat genetic diseases, advance agriculture or treat infectious diseases.



Figure 1. Gene expression. The process of transcription generates mRNA from a gene. mRNA undergoes translation to generate a protein. A change in the DNA sequence can alter the amino acid sequence of the protein and therefore can change or stop the function of the protein.

The first step in working with nucleic acids (DNA and RNA) is to remove the molecules from inside the cell. Different types of cells need to be processed differently in order to release nucleic acids. All cells have a **cell membrane**, a phospholipid bilayer that separates the internal environment of the cell from the external environment. In eukaryotes, DNA is housed inside the nucleus of the cell which is surrounded by the **nuclear membrane**, a second double-layered membrane, also composed largely of lipid molecules. When extracting DNA from plant cells, the **cell wall** must also be considered; some types of plant tissue require grinding or flash-freezing in order to break the tough cell wall.

In the DNA extraction procedure, plant cell walls and cell membranes are broken down by blending or mashing the cells. There are two key ingredients in the DNA extraction buffer aside from the water: **dish soap (detergent)** and **salt**. The **detergent** in the extraction buffer solution acts to dissolve phospholipids that form the cell membrane and the nuclear membrane. When the phospholipid membranes are dissolved, the cell lyses and releases the cellular contents, including DNA. The **salt** has two functions in the extraction process. It helps to neutralize the charge on the sugar-phosphate backbone, making DNA *less soluble* in water and allows it to more easily precipitate when the alcohol is added. The salt also helps to remove the proteins that are bound to the DNA and to keep the proteins dissolved in the lysis solution. The technique of filtration uses a medium, in this case, cheesecloth, to separate solids from liquids. The resultant material is referred to as filtrate. When cold **ethanol** is added to the filtrate, DNA precipitates at the water/ethanol interface. Although an individual DNA molecule is not visible with the naked eye, DNA isolated from large quantities of cells can be observed.

This protocol for extraction of DNA is based largely on the principle of **solubility**. Solubility refers to the ability of one substance (the solute) to dissolve in another substance (the solvent). Polar substances dissolve easily in polar solvents, but do not dissolve easily in nonpolar solvents, a phenomenon commonly referred to as “like dissolves like.” Water is a polar solvent, and molecules that dissolve easily in water are referred to as **hydrophilic**. DNA molecules are hydrophilic because the sugar-phosphate backbone of the molecules is highly polar. This means that DNA dissolves in water, so in this experiment, the DNA that is released when the cells are crushed dissolves in the juice/extraction buffer mixture.

Although the chemical reactions described above are all happening when you add the buffer and crush the food (strawberries), they are not visible with the naked eye. However, the addition of the cold ethanol caused a much more dramatic result! Ethanol is a nonpolar solvent, and when it is added to the juice extract, the DNA precipitates out of the solution. A precipitation reaction is a chemical reaction that causes a solid substance to emerge from a liquid solution. In this experiment, the addition of ethanol to the reaction forces the DNA to precipitate out of solution, which we can then spool onto the wooden stirrer or glass rod.

Exercise 1: DNA Extraction

Materials needed

1. Isopropyl alcohol 91% (rubbing alcohol) or 95% ethanol, Chilled in the freezer

2. Graduated cylinders
3. Salt
4. DI Water
5. Cheesecloth
6. Dishwashing liquid (preferably, Dawn)
7. Wide-mouth glass test tubes
8. Funnel
9. Disposable plastic cups
10. Microcentrifuge tubes
11. Wooden stirrer or glass rod
12. Resealable plastic bag
13. 100 or 200 mL beaker
14. Ice buckets with ice
15. A DNA source (about 1 cubic inch of food)
 - a. strawberries
 - b. bananas
 - c. ground flax seed
 - d. ground wheat germ
 - e. peas
 - f. broccoli
 - g. spinach

Preparation

- Have isopropyl alcohol (rubbing alcohol) or ethanol cooled in an ice bucket • Prepare your food item if needed (i.e. remove green strawberry tops, peel banana, grind wheat germ, etc)
- If not already made, prepare DNA extraction (lysis) buffer by combining the following: • 45 mL DI water
- 5 mL liquid dish soap
- 0.75 g NaCl (table salt)

Procedure

1. Place the food item (i.e 2 large strawberries) into the plastic zipper bag. Seal it and gently smash the food with your hands for about 2 minutes. Completely crush the food (strawberries) to disrupt the cells.
2. Open the plastic zipper bag and add 10 mL of the DNA extraction (lysis) buffer. Squeeze the bag to remove all air and seal the bag tightly.
3. Gently (to prevent over-sudsing or excessive foaming), but thoroughly, continue to crush the food (strawberries) inside the bag for about one minute or until it is a slushy consistency.
4. Completely line the funnel with a layer of cheesecloth. Place the funnel into the wide-mouth test tube.
5. Pour the food juice/DNA extraction buffer mixture into the funnel so the juice passes through the cheesecloth and into the test tube. Use the cheesecloth to strain the mixture so that only the juice flows into the tube and the pulp is retained in the cheesecloth.
6. Discard the cheesecloth and the pulp. Remove the funnel from the tube. The glass tube now contains a liquid called “filtrate”.
7. Carefully and slowly pipette an equal volume of ice-cold ethanol on top of the filtrate in the test tube using the plastic transfer pipette. The alcohol is less dense than the filtrate and will float as a layer on top of the filtrate. Do not mix or stir!
8. Hold the tube still at eye level and observe what happens at the interface of the alcohol and the filtrate. DNA will precipitate at the alcohol-lysis buffer interface. This means it will come out of solution into a “solid” form and appear as fluffy white cotton or cloudy material. Verify with your instructor that you have isolated DNA.
9. (optional) Use your wooden stirrer or glass rod to transfer your extracted DNA into a microcentrifuge tube. Add a small amount of ethanol to the tube to prevent your DNA from drying out.

Questions for Review

1. What reagent in the extraction buffer functions to dissolve membranes?
2. What reagent in the extraction buffer precipitates DNA so that it can be seen with the naked eye?
3. What reagent in the extraction buffer functions to remove proteins from DNA?

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4.3: Analyze Ingredient Label for GMO Ingredients

Biotechnology is the use of biological agents for technological advancement. Biotechnology was used for breeding livestock and crops long before the scientific basis of these techniques was understood. Since the discovery of the structure of DNA in 1953, the field of biotechnology has grown rapidly through both academic research and private companies. The primary applications of this technology are in medicine (production of vaccines, antibiotics and treatments and the detection of diseases) and agriculture (genetic modification of crops, such as to increase yields or make crops resistant to pests or pesticides/herbicides). Biotechnology also has many industrial applications, such as fermentation, the treatment of oil spills, and the production of biofuels.

The ability to add or modify genes in an organism is powerful technology. **Recombinant DNA technology** involves using laboratory methods to join together DNA from multiple sources or organisms. **Genetic engineering** is the alteration of an organism's genotype using recombinant DNA technology to modify an organism's DNA to achieve desirable traits. The addition of foreign DNA in the form of recombinant DNA vectors generated by molecular cloning is the most common method of genetic engineering. An organism that receives recombinant DNA is called a **genetically modified organism** (GMO). The introduced DNA causes the genetically modified organism to make the foreign protein. If the foreign DNA that is introduced comes from a different species, the host organism is called transgenic. Bacteria, plants, and animals have been genetically modified since the early 1970s for academic, medical, agricultural, and industrial purposes. Biotechnology in agriculture can enhance resistance to disease, pest, and environmental stress, and improve both crop yield and quality. As reported by the FDA, in the US in 2018, about 94% of soybeans planted, 95% of canola planted, 99.9% of sugar beets (to make granulated sugar) planted, 94% of all cotton planted, and 92% of the corn planted were genetically modified organisms. More than half of the granulated sugar on grocery store shelves is made from GMO sugar beets. Therefore, GMOs make up a large portion of some common processed foods.

Exercise 2: Analyze Ingredient Label for GMO Ingredients

Materials needed

1. Food labels for analysis

Use the questions below to analyze the ingredient label that you brought to the lab.

Questions for Review

1. A genetically modified organism makes a _____ that the organism would not normally make.
2. Is your food labeled as organic?

If the answer is yes, then according to organic food standards, this food would not contain any GMO ingredients.

3. Does this ingredient label contain any components of soybean, corn or canola?
4. If the answer is yes, then according to the previous statistics, this food likely contains GMOs.
5. If your food contains genetically modified corn, what specifically does that mean about the corn, and what exactly about the corn has been changed?
6. Would eating GMO foods have an effect on human DNA?
7. Discuss both positive and negative ways that GMO crops could possibly impact the environment.

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4.4: Test Food Products for GMO Genes

A protein that has been introduced into plants as a genetically modified trait is Bt endotoxin. Bt endotoxin (Bt toxin) is a toxin that is naturally produced by the bacterium, *Bacillus thuringiensis*. European corn borers are a type of caterpillar that causes billions of dollars of loss for the corn industry every year. If Bt toxin is consumed by European corn borer larvae it causes the corn borers to die. However, the Bt toxin is quite specific and does harm insects in other orders (such as bees and beetles) or humans. Scientists have created genetically modified corn that contains the Bt toxin gene (Cry), thus allowing the corn to produce the toxin protein (Bt corn). The Bt corn expresses the Bt toxin protein (Bt corn) is therefore resistant to the European corn borer larvae, therefore preventing the loss of crops and money.

Another gene that has been introduced into crops to produce a genetically modified trait is the gene that allows plants to be resistant to the herbicide glyphosate. Glyphosate is a widely used broad spectrum herbicide that kills both broadleaf plants and grasses. Glyphosate is commonly sold under the trade name Roundup. Some bacteria naturally express an enzyme, 5-enolpyruvylshikimate 3-phosphate (EPSP) synthase (CP4 EPSP synthase), that allows plants to survive when exposed to glyphosate. This bacterial gene was introduced into plant genomes to allow the plants to express the CP4 EPSP synthase enzyme. Glyphosate will not harm a field of genetically modified crops that express CP4 EPSP synthase, but the glyphosate will effectively kill any weeds in that field.

GMO test strips, also known as lateral flow devices or immunochromatographic strip tests, allows individuals to determine if a GMO trait is expressed in a crop sample. These tests use antibodies that bind specifically to a genetically engineered protein. If the antibody in the test strip binds to the genetically engineered protein, then a color change is observed on the test strip. This color change displays as a line on the test strip and indicates that that sample contains the genetically modified protein. Test strips have been developed to detect the presence of Bt toxin protein or CP4 EPSP synthase protein in crops. Test strips have also been developed that allow one to detect the presence of glyphosate in crops. When glyphosate is present in food crops, this indicates that glyphosate has been applied to the crop and therefore the crop must be genetically modified in order to survive this herbicide treatment. You will test a food sample for the presence of Bt toxin protein, CP4 EPSP synthase protein or glyphosate to determine if the crop is a genetically modified organism.

Materials needed

1. GMO lateral flow test strips
2. Food samples for testing

Examples of available lateral test strips:

Test strips for glyphosate: <https://detoxproject.org/testing/gly...est-home-food/>

Test strips for Cry and CP4 EPSP synthase <https://www.envirologix.com/catalog/>

Test strips for Cry <https://seedburo.com/products/3182?v...13982805819447> (~\$371/100 strips)

Test Food Sample for GMO Trait using GMO Lateral Flow Test Strip

1. Prepare samples according to manufacturer's instructions
2. Test samples using GMO test strips according to manufacturer's instructions
3. Analyze the results

Questions for Review

1. Antibodies in the GMO test strips bind to what molecules in the sample crop?
2. How is the DNA of Bt corn different from the DNA of non-engineered corn?

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4.5: Gene Therapy

Gene therapy is a genetic engineering technique used to cure disease. In its simplest form, it involves the introduction of a good gene at a random location in the genome to aid the cure of a disease that is caused by a mutated gene. In theory, the introduction of a good gene will allow the organism to produce a functioning protein instead of the mutated or absent version caused by a disease. Gene therapy attempts have often involved introducing the good gene into diseased cells as part of a vector transmitted by a virus. The virus vector (containing the good gene) can then infect the host cell and deliver the foreign DNA to the diseased cells (Figure 2).

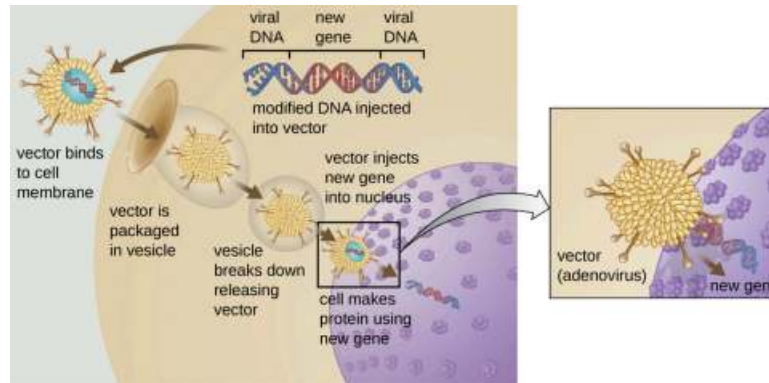


Figure 2: Gene therapy using an adenovirus vector can be used to cure certain genetic diseases in which a person has a defective gene. (credit: OpenStax Microbiology 4.4. CC-BY.)

CRISPR/Cas9

More recently, there has been much interest in correcting mutated genes using CRISPR/Cas9 technology that was discovered in bacteria. CRISPR stands for Clustered Regularly Interspaced Palindromic Repeats sequences that exist in bacterial chromosomes. Bacteria insert foreign DNA into the CRISPR sequence and use that as a template or guide to survey the cell for that foreign DNA sequence and then cleave the foreign target DNA using the Cas9 enzyme. Thus, the CRISPR/Cas9 system is a form of bacterial immune system to recognize and cleave invading DNA from bacterial viruses.

CRISPR/Cas9 involves two essential components: a guide RNA to match a desired target gene, and Cas9 (CRISPR-associated protein 9), an endonuclease which causes a double-stranded DNA break, allowing modifications to the genome (see figure 3). Cas9 is the enzyme that cuts the target DNA. Scientists have discovered that CRISPR technology makes it possible to correct errors in the genome and turn on or off genes in cells and organisms quickly, cheaply and with relative ease. One of the most exciting applications of CRISPR/Cas9 is its potential use to treat genetic disorders caused by single gene mutations. Examples of such diseases include cystic fibrosis, Duchenne's muscular dystrophy (DMD) and haemoglobinopathies. Scientists have used CRISPR/Cas9 technology to restore cystic fibrosis transmembrane conductor receptor function in stem cells from cystic fibrosis patients. Researchers have also used CRISPR/Cas9 technology to restore muscle function by recovering dystrophin protein expression in a mouse model of Duchenne's muscular dystrophy. The approach so far has currently only been validated in preclinical models, but there is hope it can soon be translated to clinical practice. Other potential clinical applications include gene therapy, treating infectious diseases such as HIV and engineering autologous patient material to treat cancer and other diseases. There has been increasing interest in the possibility of using CRISPR/Cas9 to modify patient-derived T-cells and stem/progenitor cells which can then be reintroduced into patients to treat disease, such as cancers.

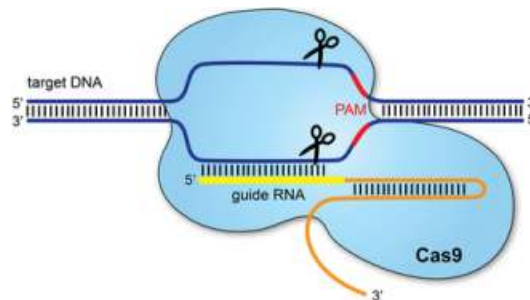


Figure 3

The CRISPR/Cas9 system. Clustered regularly interspaced palindromic repeats (CRISPR) refers to sequences in the bacterial genome. They afford protection against invading viruses, when combined with a series of CRISPR-associated (Cas) proteins. Cas9, one of the associated proteins, is an endonuclease that cuts both strands of DNA. Cas9 is directed to its target by a section of RNA. This can be synthesized as a single strand called a synthetic single guide RNA (sgRNA); the section of RNA which binds to the genomic DNA is 18–20 nucleotides. In order to cut, a specific sequence of DNA of between 2 and 5 nucleotides (the exact sequence depends upon the bacteria which produces the Cas9) must lie at the 3' end of the guide RNA: this is called the protospacer adjacent motif (PAM). Repair after the DNA cut may occur via two pathways: non-homologous end joining, typically leading to a random insertion/deletion of DNA, or homology directed repair where a homologous piece of DNA is used as a repair template. It is the latter which allows precise genome editing: the homologous section of DNA with the required sequence change may be delivered with the Cas9 nuclease and sgRNA, theoretically allowing changes as precise as a single base-pair.

The therapeutic applications of CRISPR/Cas9 investigated have predominantly been directed at somatic cells. A particularly controversial issue surrounding CRISPR/Cas9 is that of gene editing in embryos. It has already been shown that CRISPR/Cas9 technology can alter the genome of human embryos³ which theoretically could prove useful in the preimplantation treatment of genetic diseases. However, any genetic modification of the germline would be permanent and the long-term consequences are unclear. Many oppose germline modification under any circumstances, reasoning that an eventual consequence could be non-therapeutic genetic enhancement.¹³ It is clear that the ethical boundaries, within which CRISPR/Cas9 can be used, remain to be fully determined.

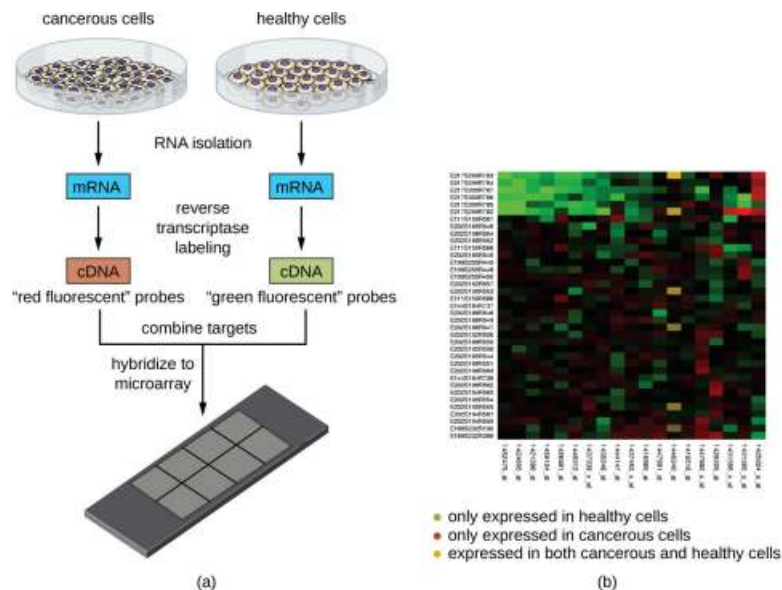
Microarrays

Some diseases, such as cancer, result from cells producing too much or too little of a protein. Microarrays are used to assess the levels of mRNA present in cells, which is indicative of the levels of proteins within cells. Microarray analysis is useful for the comparison of gene-expression patterns between different cell types—for example, cells infected with a virus versus uninfected cells, or cancerous cells versus healthy cells. Microarrays allow scientists to detect the expression of thousands of genes in one assay.

Microarray microscope slides, often called gene chips, are spotted with thousands of tiny spots containing different known DNA sequences. These DNA sequences act as probes. mRNA is isolated from cells (for example: mRNA could be isolated from a diseased cell and compared to mRNA from a healthy cell). The isolated mRNA is converted to a single stranded complementary DNA copy, cDNA, by using an enzyme called reverse transcriptase. The cDNA is labeled with a fluorescent dye and then allowed to bind to the microarray slide. cDNA will bind, or hybridize, to complementary DNA sequences on the microarray slide. Following time for hybridization to occur, the microarray slide is scanned for levels of fluorescence for each gene on the slide. Greater fluorescence associated with a gene indicates that the sample contained high levels of mRNA. No fluorescence for a specific gene indicates that the sample did not contain mRNA for that gene.

Often when performing microarrays, the cDNA from two different samples are labeled with two different fluorescent dyes (typically red and green). The fluorescently labeled cDNA samples are combined in equal amounts, added to the microarray chip, and allowed to hybridize to complementary spots on the microarray. Hybridization of sample genomic DNA molecules can be monitored by measuring the intensity of fluorescence at particular spots on the microarray. Differences in the amount of hybridization between the samples can be readily observed. If only one sample's nucleic acids hybridize to a particular spot on the

microarray, then that spot will appear either green or red. However, if both samples' nucleic acids hybridize, then the spot will appear yellow due to the combination of the red and green dyes (Figure 4). When used in this way, microarrays can generate gene expression profiles that show differences in mRNA levels of many genes when comparing control/untreated cells to diseased or treated cells.



memory immune response protects the person against that specific pathogen if it is encountered in the future. Initially, when mRNA vaccines were injected into mice they suffered severe inflammatory responses and some of the mice died. But, in 2005, Drs. Karikó and Weissman, published results showing that altering one of the mRNA building blocks, nucleosides, allowed mRNA to be less inflammatory and safely used therapeutically. It was becoming clear to their scientific team that it was possible to practically and safely use mRNA therapeutically. Early studies investigated the use of mRNA vaccines to prevent Ebola virus, but there was limited funding and demand for mRNA Ebola vaccines.

The COVID-19 pandemic brought a tremendous influx of funding and motivation to further investigate the use of mRNA vaccines to prevent or treat illness. Several different teams worked to develop mRNA COVID-19 vaccines. Extensive clinical trials showed that COVID-19 mRNA vaccines are both safe and effective. As of 2022, two independently developed mRNA COVID-19 vaccines have received full FDA approval, these are the first mRNA products to receive full FDA approval. mRNA has proven to be a powerful life-saving tool in the generation of vaccines. mRNA can be relatively easily and quickly manipulated, which makes mRNA vaccines the ideal method for the generation of vaccines against new and emerging deadly infectious diseases.

1. Sometimes children who receive the chickenpox vaccine will develop minor chickenpox blisters/rash at the sight of the injection because this vaccine contains a weakened form of the chickenpox virus (Varicella zoster virus). Is it possible for someone to contract COVID-19 infection as a result of receiving a COVID-19 mRNA vaccine? Explain why or why not.
2. Some people are concerned about mRNA vaccines modifying human DNA within the nucleus. Does this happen? Explain why or why not.
3. When the COVID-19 mRNA vaccine enters our cells it is translated to make the viral spike protein. Does this mean that our cells will make the spike protein for the rest of our lives?
4. How are mRNA vaccines considered to be safer than some other types of vaccines?

Questions for Review

1. Detergent is added to DNA extraction buffer to _____.
2. Why does DNA precipitate when alcohol is added to the DNA extraction buffer?
3. In genetic engineering, scientists make a specific change in the _____ of an organism.
4. The CRISPR/Cas9 system originated from what organisms? What purpose does this system serve in these organisms?
5. Microarrays detect differences in levels of _____ in organisms.

Testing Understanding Post-Lab

1. Sickle cell anemia is a genetic disease caused by a mutation in a single gene. This mutation causes the hemoglobin protein to be misfolded and therefore not able to carry as much oxygen throughout the body. Explain how CRISPR/Cas9 could possibly be used therapeutically to treat this genetic disease?

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4.6: References

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CHAPTER OVERVIEW

5: Lab 4 - Genomic Analysis Lab

[5.1: Overview and Purpose](#)

[5.2: Genome](#)

[5.3: DNA Gel Electrophoresis](#)

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5.1: Overview and Purpose

Overview and Purpose

This lab is designed to introduce students to the principles of genomics, DNA analysis, and data interpretation. Through hands-on lab activities and group discussion, students will gain practical skills and a deeper understanding of how genomics impacts various industries and society at large.

Learning Outcomes

1. Describe how genomes can be used in science and society
2. Explore and summarize the social and ethical issues around the use of genomes
3. Conduct and interpret DNA Gel Electrophoresis



Learning Objectives

- Run and analyze a DNA Electrophoresis Gel
- Read how genomes are utilized
- Discuss in groups the social and ethical issues of genomics

Vocabulary

DNA

Gene

Genome

Genomics

Human Genome Project

Genetic disease

Genetic testing

Gel electrophoresis

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5.2: Genome

Background Information (Read BEFORE Lab)

What is a genome?

A genome is the complete set of genes or genetic material present in a cell or organism.

Your genome is the operating manual containing all the instructions that helped you develop from a single cell into the person you are today. It guides your growth, helps your organs to do their jobs, and repairs itself when it becomes damaged. Your genome is unique to you. The more you know about your genome and how it works, the more you'll understand your own health and make informed health decisions.

Does my genome determine everything about me?

Not entirely. Genomes are complicated, and while a small number of your traits are mainly controlled by one gene, most traits are influenced by multiple genes. On top of that, lifestyle and environmental factors play a critical role in your development and health. The day-to-day and long-term choices you make, such as what you eat, if you smoke, how active you are, and if you get enough sleep, all affect your health. The way you live influences how your genome works.

Genetic Diseases

Changes in the DNA sequence are called genetic variants. The majority of the time, genetic variants have no effect at all. However, sometimes, the effect is harmful: just one deletion or alteration of just one nucleotide may result in a damaged protein, extra protein, or no protein at all, with serious consequences for our health. Additionally, the passing of genetic variants from one generation to the next helps to explain why many diseases run in families, such as in sickle cell disease, cystic fibrosis, and Tay-Sachs disease.

Genetic Testing

Predictive testing is for those who have a family member with a genetic disorder. The results help to determine a person's risk of developing the specific disorder being tested for. These tests are done before any symptoms present themselves.

Diagnostic testing is used to confirm or rule out a suspected genetic disorder. The results of a diagnostic test may help you make choices about how to treat or manage your health.

Pharmacogenomic testing tells you about how you will react to certain medications. It can help inform your healthcare provider about how to best treat your condition and avoid side effects.

Reproductive testing is related to starting or growing your family. It includes tests for the biological father and mother to see what genetic variants they carry. The tests can help parents and healthcare providers make decisions before, during, and after pregnancy.

Direct-to-consumer testing can be completed at home without a healthcare provider by collecting a DNA sample (e.g., spitting saliva into a tube) and sending it to a company. The company can analyze your DNA and give information about your ancestry, kinship, lifestyle factors and potential disease risk.

Forensic testing is carried out for legal purposes and can be used to identify biological family members, suspects, and victims of crimes and disasters.

Genomic impact on everyday life

Ethical and Social Questions: The scientists who launched the Human Genome Project recognized immediately that having a complete human genome sequence would raise many ethical and social issues. In 1990, the Ethical, Legal, and Social Implications (ELSI) Research Program was formally established at the National Institutes of Health (NIH) as an integral part of the Human Genome Project. The research supported by this program ranges from genomics and health disparities to inclusion of diverse populations in genomics research, to whether people should have the right to refuse to know genomic testing results.

Consent and Privacy: Among the major areas of study in ELSI research are questions about consent and privacy. As new areas of genomics have developed in recent years (like learning about microbiomes), researchers have needed to continually update their

guidelines, so as to help people understand the relevant risks and benefits before signing up to be a research participant. Such studies are overseen by Institutional Review Boards (or IRBs), and these boards are made up of scientists, ethicists, and members of the community. An IRB must approve any research projects involving humans.

The widespread availability of genomic data has brought changes to privacy considerations as well. When a test is performed on your DNA how the privacy of your genomic data is handled is a concern for many. Since you share half of your genome with each of your parents, and half with each of your children, the information is not just about your genome.

Answer the following questions before moving on.

Let's say that one of your parents learns from a genetic test that they have Huntington's disease, which is often diagnosed quite late in life. This gives you a 50% chance of carrying the same genetic mutation for this fatal neurological disease. Some people react to such information by wanting to know right away what their future might be, while others do not want to know.

STOP AND THINK.

Take a moment to answer the following question.

Should you be able to stop your relatives from revealing genomic information that could be relevant to you as well?

Discrimination: Concerns also exist about potential discrimination based on information about a person's DNA. In 2008, President Bush signed into law the Genetic Information Nondiscrimination Act (or GINA for short). This law is now enforced by multiple federal agencies, including the United States Equal Employment Opportunity Commission. GINA also prohibits health insurers from discriminating against anyone based on their genetic information when determining coverage or rates.

Direct-to-Consumer testing: Advances in genomics since the Human Genome Project have led to an explosion in at-home, or "direct to consumer" genomic testing.

There are now numerous companies offering you a service to learn more about yourself or your family through DNA testing. The global "direct-to-consumer" (or DTC) genomic testing market was valued at \$117 million in 2017, suggesting millions of people are purchasing these tests.

One popular application of such DTC genomic testing is for revealing family ancestry or genealogy. For example, the company Ancestry.com started as a place to search historical and family records to gain genealogical clues, and has now expanded to offer genomic tests for people who seek information about their genomic ancestry.

Another prominent company, 23andMe, was established in 2006 with the goal of offering genomic testing directly to people interested in getting DNA-based health, traits, and ancestry information. Some of the information reported back to customers are surprising facts, such as whether you have dry or wet earwax due to genomic variants or if you are likely to prefer salty/savory snacks compared to sweets. They also offer their customers the opportunity to provide their genomic information to large research studies, such as recruiting 25,000 people for a study on the genomics of schizophrenia and bipolar disorder. These research studies can provide another avenue to advance genomic science and improve human health on a rapid basis.

In recent years, there have been a number of stories involving people who found new family members or famous relatives through DTC genomic testing. Some people have found out that family members who they previously thought were genetic relatives were actually not biologically related to them, or even that their own identity is not what they thought.

Let's take one example. Some DTC genomic testing companies claim to provide information about a child's ability in certain sports. You want to find out if your daughter will be a sprinting champion. There is validated science behind the claim that certain variants in the gene ACTN3 control different types of muscle fibers (called "fast-twitch" or "slow-twitch"), and many successful sprinters have the "fast-twitch" genomic variants. But the prediction power stops there. ACTN3 variants are just one factor among many, many others - genomic, environmental, and social. So, even if your daughter does not have the "fast-twitch" genomic variant, that does not mean she cannot become an accomplished sprinter. Looking for genomic variants in children, such as the one for "fast-twitch" muscle fibers, also raises ethical questions.

STOP AND THINK

Answer the following question.

1. Since children may not be able to consent to such a test and/or they might be unable to fully understand what such testing might mean for them today or in their future, is it okay for parents to have their children tested? Might you treat your child differently based on the results?

Pharmacogenomics: Pharmacogenomics is a branch of pharmacology concerned with using DNA and amino acid sequence data to inform drug development and testing. An important application of pharmacogenomics is correlating individual genetic variation with drug responses.

Agricultural: Genomics enables farmers to accelerate and improve plant and animal breeding practices that have been in use for thousands of years. The ability to read genome sequences coupled with technologies that introduce new genes or gene changes allows us to speed up the process of selecting desirable traits in plants and animals.

Genome modification: The ability to read genome sequences coupled with technologies that introduce new genes or gene changes now allow people to speed up the ability to select for desirable traits in plants and animals. By mimicking natural processes, scientists can selectively add traits like resistance to herbicides in plants. The resulting offspring have been called genetically modified organisms (or GMOs). One example is "Golden Rice," which is a rice strain that has small bits of corn and bacterial DNA added to its genome. These extra genes allow the rice to produce beta carotene (a vitamin A precursor). The lack of vitamin A affects millions in Africa and Asia, causing blindness and immune system deficiencies.

Food Safety Monitoring: One of the most important agricultural advances in the 20th century has been the ability to move food around the globe to people who need it. Unfortunately, food supplies sometimes have unwanted guests along for the ride, such as bacterial pathogens. When people eat contaminated food, they can get very sick or even die, so it's important to find the pathogens and eliminate them. The U.S. Food and Drug Administration (FDA) has an entire

network set up for whole genome sequencing of bacterial contaminants in food, called GenomeTrakr. In 2017, this database had over 5800 bacterial sequences added on average each month, as scientists tracked new outbreaks in the quest to keep our food safe. FDA scientists work closely with others from the U.S. Centers for Disease Control and Prevention, the U.S. Department of Agriculture's Food Safety and Inspection Service, and state health departments to identify bacteria that might cause outbreaks from food contamination. Rounding out this network, the National Center for Biotechnology Information keeps track of what foods are linked to each incident, as well as in human patients who got sick. These information sources have been crucial for lowering the impact of foodborne illnesses over time.

Human Origins and Ancestry: Advances since the Human Genome Project allow us to compare genome sequences among humans, living and long-deceased, and to trace our collective ancestral history.

Species History: Nearly 20 years ago, scientists developed techniques for extracting small amounts of DNA from ancient samples, like bones or fur or even soil, and used very sensitive methods for sequencing the extracted DNA. Genomic studies like these have allowed us to examine human genomes from around 500,000 years ago when our ancestors (the species *Homo sapiens*) were diverging from other similar species, such as *Homo neanderthalensis* or Neanderthals.

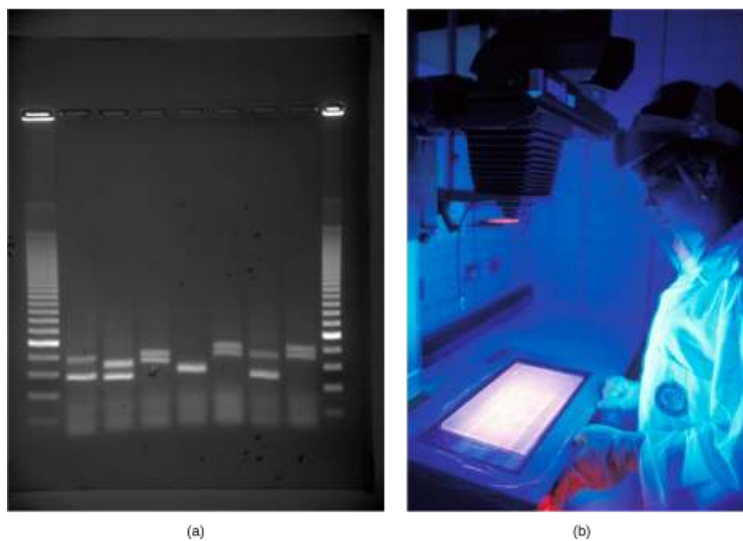
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5.3: DNA Gel Electrophoresis

One of the commonly used techniques to compare genomes is DNA gel electrophoresis. This technique is used to separate DNA fragments of different lengths resulting from a restriction enzyme digestion (using an enzyme to cut the DNA, essentially a DNA scissor). It takes advantage of the fact that a DNA molecule, or fragment, has an overall negative charge, and is thus attracted to a positive electric field. In addition to a digested DNA sample, DNA gel electrophoresis requires a gel electrophoresis chamber, which consists of a water-tight apparatus, with a positive electrode on one end and a negative electrode on the other, containing an agarose gel.

The agarose gel, which is typically the shape of a thin slab, is a semi-solid gelatin matrix that has tiny holes that DNA fragments can move through. The gel is placed into the chamber with the wells closest to the negative electrode. The chamber is then partially filled with a clear electrophoresis buffer, which contains ions that allow it to carry electrical current. The gel has several empty spaces on one end called loading wells into which the DNA samples are loaded. Once the DNA samples are loaded into the wells, the chamber is then connected to a power supply and voltage is applied. Because of DNA's negative charge, fragments will migrate through the gel toward the positive end of the chamber. The smaller the DNA fragment, the faster it will migrate (run) through the gel matrix, resulting in the separation of the fragments based on size. Once electrophoresis is complete, the smallest DNA fragment in each well will be

closest to the positive electrode, and the largest DNA fragment will be closest to the loading well (the negative electrode). DNA fragments of the same size will collect at the same location in their lane, creating a band that can be seen. Any given band within a gel may have millions of DNA fragments of equal length.



a) Shown are DNA fragments from seven samples run on a gel, stained with a fluorescent dye, and viewed under UV light; and b) a researcher from International Rice Research Institute, reviewing DNA profiles using UV light. (credit: a: James Jacob, Tompkins Cortland Community College b: International Rice Research Institute)

Materials needed

1. Micropipette
2. Pipette tips
3. Gel electrophoresis chamber
4. Agarose gel
5. Practice loading gel
6. Distilled water
7. Pigment samples
8. Electrophoresis buffer
9. Power Supply

Activity

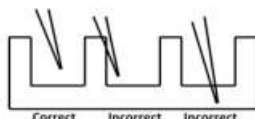
In this lab activity you will do the following

- Load a agarose gel with various samples using a micropipette
- Run the gel in an electrophoresis chamber
- Analyze the results

Practice Gel Loading

When working with DNA samples, you are measuring and dispensing volumes in microliters (μl). In order to accurately measure such tiny volumes requires a special device called a micropipette. Prior to working with your actual samples and your actual agarose gel, you will practice with the loading dye and a 'practice gel'. After watching a demonstration of how to use a micropipette and load a gel, perform the following:

1. Cover your practice loading gel with water (to simulate gel electrophoresis buffer).
2. Load a tip onto your micropipette.
3. Draw up $10\mu\text{l}$ of practice loading dye:
 - a. With your thumb, depress the micropipette plunger down to the first stop.
 - b. Place the micropipette tip into the practice loading dye.
 - c. Draw up practice loading dye by slowly releasing the plunger.
 - d. Pull the tip out of the practice loading dye—you now have exactly $10\mu\text{l}$ of dye in the micropipette tip.
4. Carefully load the practice loading dye into a well of the practice loading gel:
 - a. Place the tip of the micropipette into the well, being careful not to pierce the bottom of the gel with the tip (this would cause your sample to 'leak' out of the bottom of the gel).
 - b. Dispense the dye into the well by slowly depressing the plunger to the second stop.
 - c. Without releasing the plunger, pull the tip out of the well (if you release the plunger too soon, you will suck your sample back up).
 - d. After the plunger is out of the well, slowly release the plunger.

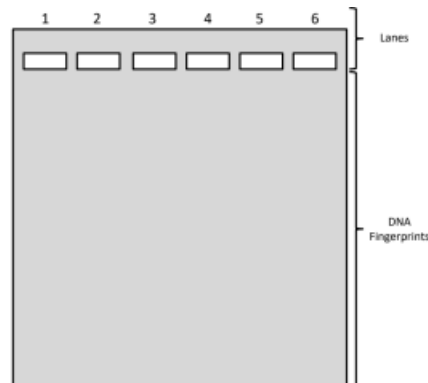


5. Each group member should repeat steps 3-4 several times so that everyone is comfortable using a micropipette and loading a gel.
6. When finished, carefully remove the tip from the micropipette and dispose of it in the biohazard container.
7. Rinse your practice gel with cold tap water to remove the dye and return it to your table.

Gel Electrophoresis Activity:

1. Take the agar gel that has been set and solidified.
2. Remove the cover of the chamber by sliding it off while holding the chamber firmly.
3. Place the gel in the center of the electrophoresis chamber, oriented such that the wells are closest to the negative (black) electrode of the chamber.
4. If necessary, add more electrophoresis buffer to the chamber so that the top of the gel is covered by approximately 1cm of buffer.
5. Carefully load $10\mu\text{l}$ of each sample into the wells of the gel.
6. Place the cover on the gel electrophoresis chamber and connect the electrode terminals to the power source.
 - a. Make sure that the negative (black) and positive (red) color-coded indicators on the cover and chamber are properly oriented.
 - b. Likewise, make sure that the electrodes are plugged into the correct electrode terminals on the power source. Before proceeding to the next step, ask the instructor to inspect your setup.
7. Plug in and turn on the power source and set it to 75V. You should be able to see bubbles forming at each end of the chamber.

8. Run your gel for approximately 60 minutes.
 - a. Caution: do NOT allow your gel to run too long, or else the fragments will literally run out of the end of the gel!
 - b. Carefully monitor the progress of your samples every 5 minutes.
 - c. Typically, the gel is done and should be turned off when the blue 'fast dye' has run approximately 2/3rds down the gel.
 - d. When you think the gel has finished running, let your instructor know.
9. After electrophoresis is complete, turn off and unplug the power source, then carefully remove the cover from the chamber.
10. Using the figure below, draw a picture of your results (using color pencils) and answer the questions.
11. Follow your instructor's instructions on how to clean and reset your gel electrophoresis experiment.



Questions

1. What would happen if you positioned the agarose gel backwards within the chamber, with the sample wells on the positive electrode end of the electrophoresis chamber?
2. Describe why larger (longer) fragments move through an agarose gel more slowly than smaller (shorter) fragments?

Group Discussion Activity

Your professor will separate you into small groups. You will be given a genomic scenario to discuss and questions to answer. The following are the possible scenarios.

Scenario 1: Amy and the breast cancer test

This scenario deals with a 15 year old girl (Amy) who has a family history of breast cancer. Her paternal aunt has been diagnosed with the disease and tests have shown that she is a carrier of the BRCA1 mutation associated with breast cancer. There is a dispute within the family about whether Amy's father should be tested to find out whether he is also a carrier of the BRCA1 mutation, and may have passed the risk to Amy. Her mother does not want the test to take place. The decision is Amy's. The group has to decide whether Amy should ask her father to take the test.

Scenario 2: Jill and the schizophrenia risk

This scenario is set in the future and assumes that people will be able to find out this information through a recreational genotyping kit. Summary: Scenario 2 deals with a woman (Jill) who has no knowledge of her father's family history. She has a husband and a young baby and pays to be genotyped to get a better understanding of her genetics. She does not know that the test will reveal that she has a rare chromosome deletion, associated with a higher than normal risk of developing schizophrenia.

Scenario 3: Sam and the insurance company

This scenario is set in the future and assumes that insurance companies will be able to use genetic information to determine insurance premiums. Summary: Scenario 3 deals with a middle-aged man (Sam) with a family history of heart attacks. He wishes to purchase insurance but the insurers ask him to be genotyped. Sam's brother (who has had heart problems himself) is keen for Sam to be tested. He feels that knowing if Sam is at high risk of heart disease will help him to take steps to avoid severe heart problems. Sam fears he will be refused insurance if he takes the test and is not certain he wants to know the results. His wife is firmly against genotyping. The group has to decide if Sam should take the test.

Scenario 4: Heather, her risk of Alzheimer's and her reluctant twin

This scenario is set in the future and assumes that recreational genotyping has become commonplace. Summary: This scenario presents a young woman (Heather) who has an identical twin. Heather wishes to start a family with her husband and has decided to get herself genotyped. Her twin objects to the test and does not wish to know the results. The results show that the twins are carriers of two copies of the gene variant ApoE-e4, which puts them at high risk of developing Alzheimer's disease. The group has to decide whether Heather should tell her sister or parents about the results. As the twins are identical, whatever is revealed about the genotype for one twin will match the genotype of the other. If the twins were non-identical the genetic similarities would only be as close as that of any other pair of siblings.

Scenario 5: Pete's potential adverse drug reaction

This scenario is set in the future. A test for the genes linked to this drug reaction does currently exist, but is not currently widely available to patients. Summary: This scenario is centered on (Pete) whose doctor would like to prescribe the antibiotic flucloxacillin, since it is particularly effective for his type of infection. However, in rare cases flucloxacillin can cause a serious liver injury, and Pete is offered a test to determine whether he has a particular genotype associated with this severe reaction. Pete is concerned about the cost but his wife is keen for him to take the safe option. His teenage son is concerned about misuse of the data if it becomes part of his medical records. The group has to decide whether Pete should take the test.

Scenario 6: Should a baby have its genome sequenced?

This scenario is set in the future. It assumes that we are at a point where whole genome sequencing can be carried out cheaply and quickly. Summary: In this scenario expectant parents are asked whether they would like to have their child's genome sequenced after it is born. The information would form part of the child's ID card.

Scenario 7: Andy's unexpected paternity results

This scenario is set in the future and assumes that recreational genotyping has reached the point that it is affordable for most people to purchase a genotyping kit as a gift. Summary: This scenario involves Andy, who buys a genotyping kit online. He finds it fun and therefore decides to buy one for his father. Looking through their results together, Andy notices that there are different markers on the Y chromosomes of him and his father, indicating that he is not Andy's biological father. The group has to decide whether Andy should tell his father what he's found.

Scenario 8: Should researchers share incidental findings?

This scenario is set in the present day. Summary: This scenario is about a researcher (Lin) who is working on a study into childhood developmental disorders. She discovers that one of the children in the study has a mutation associated with the cancer retinoblastoma. The study has a strict policy not to share incidental findings with participants. The group has to decide whether Lin should tell the child's parents.

For review only as this link will not be included in the final draft.

[CLICK HERE!](#) To view the supporting materials for the listed scenarios

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CHAPTER OVERVIEW

6: Lab 5a - Investigating Fossils - Understanding the Past through Paleontology

[6.1: Overview and Purpose](#)

[6.2: Fossils](#)

[6.3: Data Analysis](#)

[6.4: Conclusion](#)

This page titled [6: Lab 5a - Investigating Fossils - Understanding the Past through Paleontology](#) is shared under a [CC BY 4.0](#) license and was authored, remixed, and/or curated by .

6.1: Overview and Purpose

Overview and Purpose

This lab is intended to provide a hands-on learning experience about the science of paleontology and the process of fossilization. Through examining and analyzing actual fossils, students will gain an understanding of how fossils are formed and the significance of these remains for reconstructing the past.

Learning Outcomes

1. Define what fossils are and how they provide insight into the Earth's past.
2. Outline the general timeline of life on Earth.
3. Describe of the timescale relationship of the Universe, the Earth, and life.



Learning Objectives

- Make observations and take measurements of fossils.
- Draw comparisons between fossils and their ages
- Understand the position of life in geologic and cosmic time.

Vocabulary

Fossil

Mummification

Geologic time

Paleozoic

Mesozoic

Cenezoic

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6.2: Fossils

Background Information

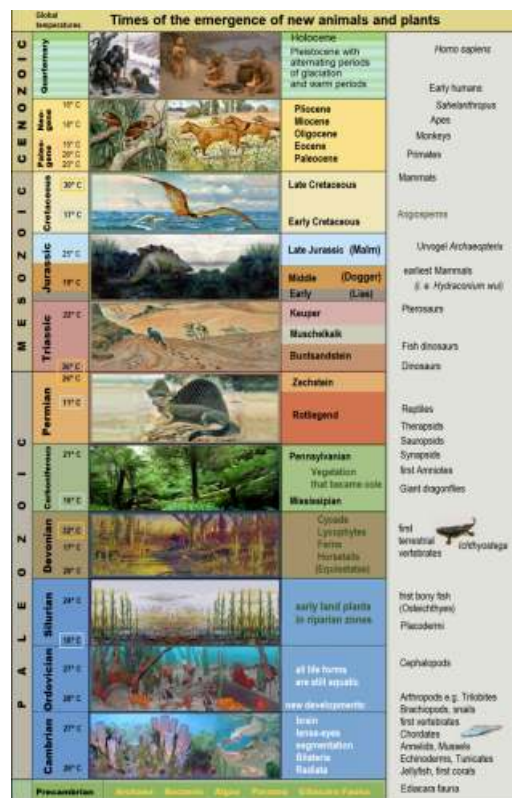
The history of life on Earth began around 3.5 billion years ago with the emergence of single-celled microorganisms, called prokaryotes. Over time, these evolved into more complex cells, including eukaryotes, which eventually led to the formation of multicellular organisms. Around 540 million years ago, the first complex life forms appeared in the oceans, including invertebrates and early fish. During the next 400 million years, life continued to evolve and diversify, leading to the emergence of land plants and animals, including dinosaurs. The extinction of dinosaurs 65 million years ago paved the way for the rise of mammals, including primates and eventually humans. Over the past 200,000 years, humans have developed complex civilizations, languages, and technology.

A **fossil** forms when an organism's remains or imprint are preserved in rock or other material. This typically happens when the remains are covered in sediment, such as sand, ash, or mud, which eventually hardens into rock. Over time, the organic material decays, leaving behind an impression or mold in the rock. In most cases, minerals replace the organic material, preserving it as a mineral cast. Fossils can also form from preserved tracks or burrows made by an organism, or from preserved body parts, such as leaves, shells, or bones. The process of fossilization can take millions of years, and the conditions must be just right for preservation to occur.



FOSSIL CRINOID. A MARINE ANIMAL, FLOWERLIKE, ANCHORED BY A STALK. GRCA 14466. 26 MAR 1993. QUINN, NPS. CCBY 2.0

Soft tissues, such as muscles, organs, and skin, are not typically preserved in fossils. However, in rare cases, conditions may exist that allow for the preservation of soft tissues. One way this can happen is through rapid burial, which helps prevent decay and the destruction of soft tissue by scavengers. Another way is through the process of mummification, which occurs in dry, arid environments where low humidity slows down decay. In recent years, researchers have discovered many examples of fossilized soft tissues, such as dinosaur skin and feathers, body impressions, stomach contents, as well as many other exciting things. These fossils provide important insights into the anatomy and biology of ancient organisms.



Evolutionary history of life. Montage by Scienica58 is based on works by Richie Diesterheft, JJ Harrison (jjharrison89@facebook.com), Ghedoghedo, Fritz Geller-Grimm and others., CC BY-SA 4.0
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Materials needed

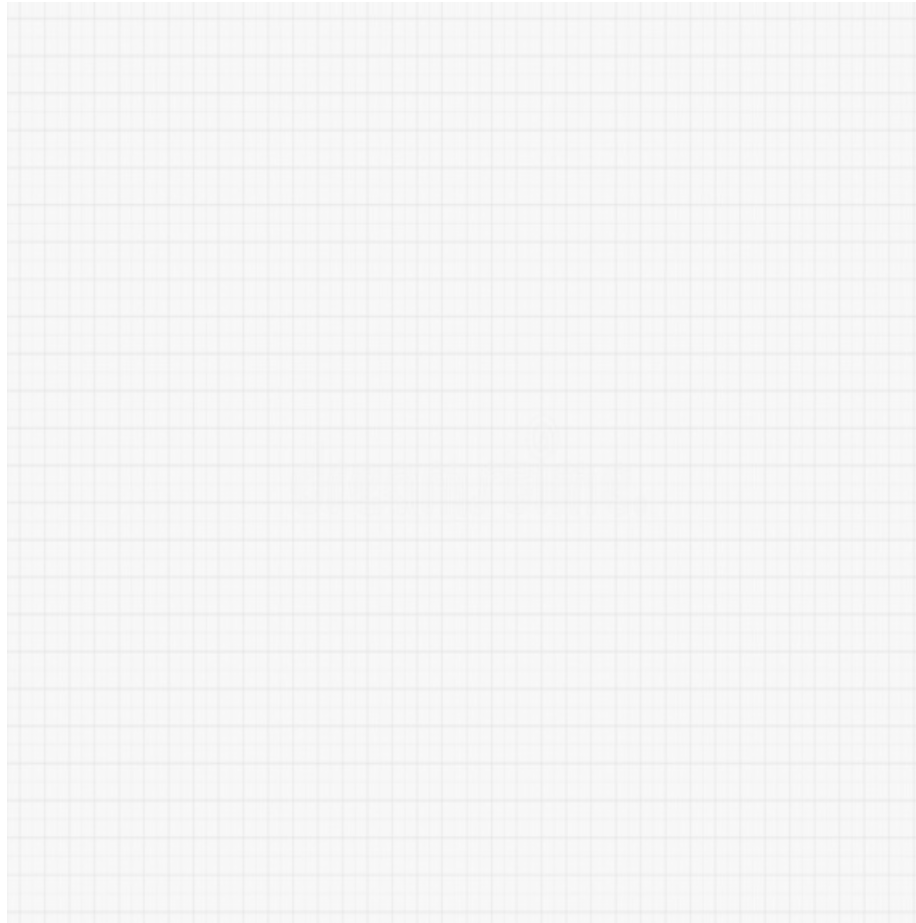
1. Fossil specimens (real or replicas)
 - a. 4-6 specimens per 3 students
2. Magnifying glass or dissection microscope
3. Sketchbook and pencil
4. Reference materials (books, websites, etc.)
5. Rulers
6. Device with photographing capability
7. Device with internet access

Procedure:

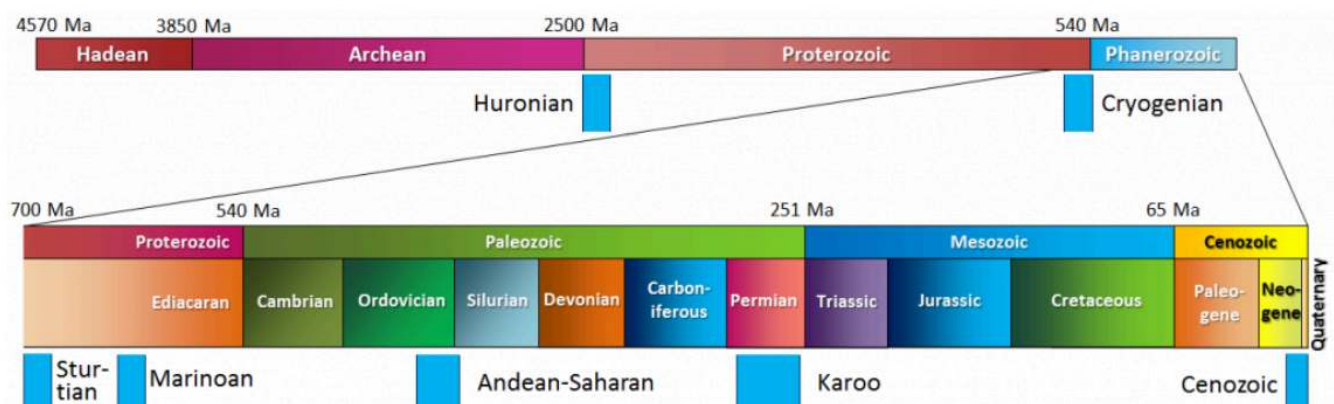
1. Obtain a fossil specimen from your instructor. There should be some basic information included with it such as where it was found, type of rock, the general age of rock, and a general description of the species. Also, obtain a magnifying glass or other magnifying device.
2. Observe each fossil specimen with a magnifying glass and examine its physical characteristics. Take note of any noticeable details, such as the type of rock it is found in, any visible imprints or textures, and the overall shape of the fossil.
3. Make detailed sketches of your fossil in the space below. Be sure to include a description of what you observed, measurements, and any other observations you have made.
4. After sketching, take a picture of your fossil along with a ruler to show scale. Insert the picture beneath your sketch.
5. Research the type of fossil you have found. Use the basic information that was provided to you as a starting point. Use reference books or online resources to identify its age, the species it belonged to, and the environment in which it lived.
6. Record your observations and research in a lab report. Include sketches of the fossils and describe their physical characteristics and the environment in which they were formed. (OPTIONAL)

7. As a class, discuss the findings and how they relate to the Earth's history. 8. On the timeline and table below, correctly write in the fossils to the date.

Draw your fossil sketch and notes in this space. Include any comments or observations you make here as well.



Insert photo with scale here:



- *The record of major past glaciations during Earth's history. Source: Steven Earle (2015) CC BY 4.0 [view source](#)*

The importance of fossils in providing evidence of the Earth's past are invaluable and help us piece together the history of life on our planet.

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6.3: Data Analysis

1. Based on your research, create a chart or table comparing the different fossils you have found, including their age, species, and environment.
2. Discuss any patterns or trends that you observe in the data. For example, are the fossils all from a similar time period or from the same environment?
3. Compare your findings to what is known about the evolution of life on Earth. In the timeline above, mark the timeline with your class's fossils. What can we learn about the ancient environment and species from these fossils?

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6.4: Conclusion

1. Summarize your main findings and explain what you learned about the process of fossil formation.
 2. Discuss the significance of fossils and their importance in understanding the history of life on Earth. Can you think of any other way that we could look back in time at what happened in life's past?
 3. Explore the website [Chronozoom\(link is external\)](#). Can you get a sense of the timescale of life versus the timescale of our planet versus the timescale of the universe? Describe any observations you make.
 4. Reflect on what you have learned from this lab and what you might do differently next time to further enhance your understanding of fossils.
-

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CHAPTER OVERVIEW

7: Lab 5b - Classification and Tree Thinking

[7.1: Overview and Purpose](#)

[7.2: Taxonomy](#)

[7.3: Part 1- Dichotomous Keys](#)

[7.4: Part 2- Cladistics](#)

[7.5: Part 3- Making Caminalcule Cladograms](#)

[7.6: Lab Questions](#)

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7.1: Overview and Purpose

Overview and Purpose

Examine the principles of taxonomy and phylogenetics and use those principles to observe and describe the characteristics of different organisms. Use specimens, taxonomic keys, and guides to identify organisms, and work in teams to compare their observations and conclusions. This lab emphasizes the development of skills in observation, classification, and scientific reasoning. This lab explores the diversity of life and the importance of taxonomy and phylogeny in understanding the natural world.

Learning Outcomes

1. Understand the processes scientists use to determine the relatedness of different species.
2. Gain an understanding of dichotomous keys and cladograms.
3. Understand the differences between taxonomy and cladistics.



Learning Objectives

- Use and make a dichotomous key.
- Use and label a cladogram to understand organismal relationships, ancestral traits, and shared derived traits.
- Make a scientifically accurate cladogram using a fictional example.

Vocabulary

Taxonomy

Linnaean System

Dichotomous Key

Cladogram

Phylogeny

Phylogenetic Tree

Ancestral trait

Shared derived traits

Clade

Materials needed

1. A selection of preserved animals from a school collection or from student-made collections
2. Printouts of 'Caminacules' from: Sokal, R. R. (June 1983). "A phylogenetic analysis of the Caminalcules. I. The data base". *Systematic Zoology*. **32** (2): 159–184. doi:10.2307/2413279. JSTOR 2413279

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7.2: Taxonomy

Background Information

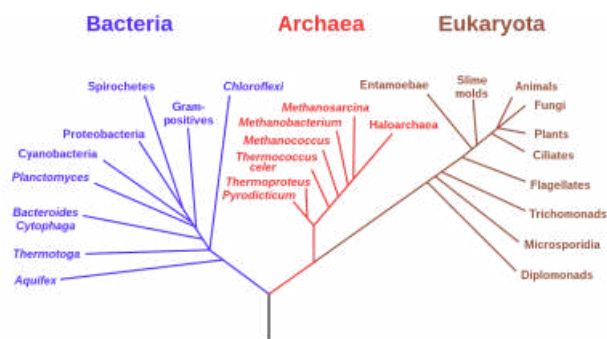
Humans are excellent classifiers. Scientists classify organisms based on features that are shared in common with other related organisms. The formal process of classifying organisms is referred to as **taxonomy**.

Traditional taxonomy relies primarily on physical traits so that organisms that look alike are placed in the same group. One traditional taxonomic system, called the **Linnaean system** of taxonomy, established the following categories:

	Critter 1	Critter 2
Domain	Eukarya	Eukarya
Kingdom	Animalia	Animalia
Phylum (Division)	Chordata	Arthropoda
Class	Mammalia	Insecta
Order	Carnivora	Hymenoptera
Family	Felidae	Formicidae
Genus	Felis	Solenopsis
Specific Epithet	domestica	invicta

Can you tell what organisms are categorized here?

Classifying organisms based on shared evolutionary history attaches powerful information to a classification scheme. While traditional taxonomy relied primarily on classifying organisms by their shared physical characteristics (or morphology), technological advancements have allowed scientists to determine an organism's evolutionary history more accurately. Taxonomy based on presumed evolutionary relationships is referred to as **phylogeny**. Phylogeny is similar to constructing a family tree in that shows how different species are related to each other the same way that you are related to your ancestors. The largest difference is that a phylogenetic tree shows the relationship between populations instead of individuals (as in a family tree). It helps us understand how different species have evolved over time and how they are related to one another. Just like a family tree, a phylogenetic tree has branches and roots, with the roots representing the earliest ancestors and the branches showing how species have evolved and changed over time. By studying phylogeny, we can learn about the history of life on Earth and the relationships between different species.



A phylogenetic tree of living things, based on RNA data and proposed by Carl Woese, showing the separation of bacteria, archaea, and eukaryotes. (SVG phylogenetic tree of life CC0)

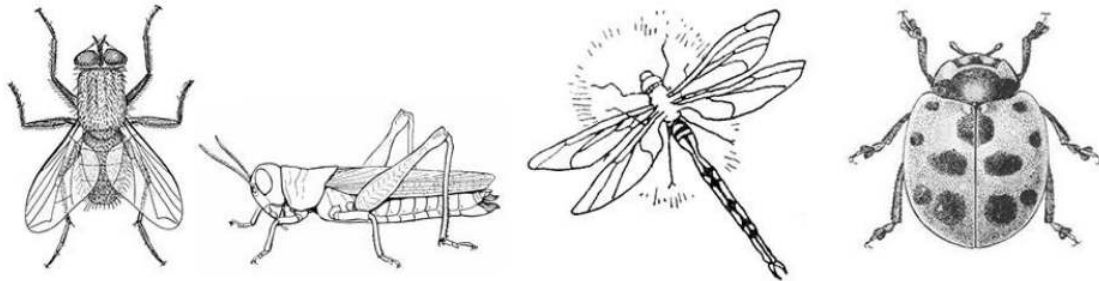
Usually, traditional taxonomic systems agree with phylogenetic systems when determining where organisms should be classified. However, sometimes, striking conflicts can arise. For example, birds are traditionally placed in their own class *Aves*. However, a wealth of research indicates that birds are actually living dinosaurs, and are thus technically reptiles.

Check out some [more information on this fascinating situation](https://evolution.berkeley.edu/learn-evolution/). More information can be found here:
<https://evolution.berkeley.edu/learn-evolution/>

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7.3: Part 1- Dichotomous Keys

A **dichotomous key** is an identification key that consists of a series of choices that ultimately lead to the identification of the object in question. In each step of a dichotomous key, the user has two choices and must pick the one that most correctly describes the object. Each choice leads to a new set of choices. To help you understand how a key like this works, consider the following four insects:



After studying the insects, you might classify the insects by wing covering, body shape, and the direction in which the wings point. To begin the key, you could start separating the four insects based on wing covering— “flight wings covered by exoskeleton” vs. “flight wings not covered by an exoskeleton.”

The first step in the key could be organized the following way:

1. wings covered by an exoskeleton. **go to step 2**
2. wings not covered by an exoskeleton **go to step 3**

Step 2 consists of a pair of statements that distinguishes between the ladybug and the grasshopper.

1. body has a round shape **ladybug**
2. body has an elongated shape. **grasshopper**

Step 3 consists of a pair of statements that distinguishes between the dragonfly and the housefly.

1. wings point out from the side of the body **dragonfly**
2. wings point to the posterior of the body. **housefly**

Notice that there were four organisms to be identified and it only took three steps to identify them. After making a key, you should end up with one less step than the total number of organisms you are trying to identify.

When constructing a key, keep the following in mind:

- Use constant characteristics rather than variable ones.
- Use measurements rather than terms like “large” and “small.”
- Use characteristics that are generally available to the user of the key rather than seasonal characteristics or those seen only in the field.
- Make the choice a positive one—something “is” instead of “is not.”
- If possible, start both choices of a pair with the same word.
- If possible, start different pairs of choices with different words.

When using a key, keep the following in mind:

- Always read both choices, even if the first seems to be the logical one.
- Be sure you understand the meaning of the terms involved. *Do Not Guess.*
- When measurements are given, use a calibrated scale. *Do Not Guess.*
- Since living things are always somewhat variable, do not base your conclusion on a single observation. Study several specimens to be sure your specimen is typical.

- If the choice is not clear, for whatever reason, try both divisions. If you end up with two possible answers, read descriptions of the two choices to help you decide.
- Having arrived at an answer in a key, do not accept this as absolutely reliable. Check a description of the organism to see if it agrees with the unknown specimen. If not, there is an error somewhere, either in the key or in its use. For a scientist, the ultimate check of identification is a comparison of the unknown with an authentically named “Type Specimen.”

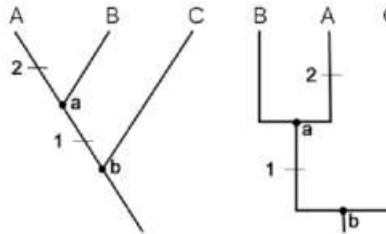
Procedure

1. Examine the 6 specimens from school collections or organisms you collected and use them to construct your own dichotomous key here (use the example above as a template):
2. Use someone else’s key to identify 2 specimens. List your path to identification here (letters are fine):
 - a. Specimen 1
 - b. Specimen 2

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7.4: Part 2- Cladistics

A newer system of phylogenetic classification, called **cladistics**, is currently in practice today. A **cladogram** is a hypothesis about the evolutionary relationships between the organisms depicted on the tree. In this way, a cladogram illustrates the lines of descent for these organisms. A cladogram proposes an answer to the question “Which groups of organisms share a common ancestry?”



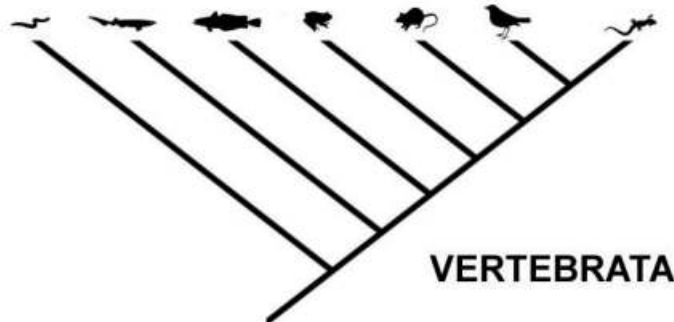
Take a look at these two identical, generic cladograms. The capital letters indicate the terminal organisms represented in the tree. The numbers indicate characters present in organisms beyond that point. In addition, the nodes (indicated by the lowercase letters and the dots) represent the common ancestors of the terminal organisms. Even though they look different, if you examine them closely, these two cladograms are depicting the same relationships between critters A, B and C. As a mental exercise, try replacing A, B, and C with 3 animals and see if you place them correctly based on relatedness. Check with your instructor if you are unsure of how they are related.

In this example, you see that A is more closely related to B than it is to C based on the shared derived characteristic 1. Note that at each branch a derived characteristic is indicated that separates the left branch from the right branch of the evolutionary tree.

Now examine the cladogram below illustrating the evolutionary relationships between a hagfish, shark, bony fish, frog, rat, bird, and lizard.

Procedure

1. Name each organism on the cladogram.
2. Place a dot at every point that represents a common ancestor.
3. Indicate one shared derived characteristic that distinguishes each branch.
4. Who is more closely related: the shark and bony fish, or the bony fish and frog?



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7.5: Part 3- Making Caminalcule Cladograms

The evolutionary biologist Joseph Caminal created **Caminalcules** as an example model system to understand phylogenetics. He imagined 29 living species and 48 fossils of these fictitious animals which were published by Robert R. Sokal (1983a) in the journal *Systematic Zoology*. They were used to conduct detailed research on evolutionary classification. (Sokal, R. R. (June 1983). "A phylogenetic analysis of the Caminalcules. I. The data base". *Systematic Zoology*. **32** (2): 159–184. doi:10.2307/2413279. JSTOR 2413279)

As an alternative, your instructor may ask you to construct a cladogram for the specimens that you used for the dichotomous key.

Caminalcule Procedure

1. As an example, list the similarities and differences among the following individuals.
2. Draw the simplest possible evolutionary tree that contains these 4 **living** critters and 3 **fossil** critters.
3. Include at least one trait that distinguishes each living critter.
4. Use each fossil as an **ancestor**.
5. Remember—cladograms are hypotheses that must be tested. Your hypothesis might be different from someone else's!

The Caminalcules

Using the provided sheet of paper with illustrated Caminalcules, cut out each individual using a pair of scissors. Next, working in groups of 2 or 3, categorize individuals into subgroupings based on shared morphological features that distinguish them from other subgroupings (known as a **shared derived feature**). Also, pay attention to traits that all Caminalcules share regardless if they are living or extinct. These are known as **ancestral traits**. Once you have categorized the species into four or five major groups, paste the animals onto a piece of paper and indicate branches delineated by a shared derived feature. The instructor will illustrate an example of a completed cladogram.

Remember! Cladograms are hypotheses that must be tested! Your hypothesis might be different from someone else's. Be prepared to *defend* your decision with evidence to someone whose hypothesis is different.

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7.6: Lab Questions

1. What is the difference between cladistics and taxonomy?
2. How does cladistics help in classifying species?
3. What is a clade in cladistics?
4. How does cladistics determine the evolutionary relationships between species?
5. Can cladistics be used in combination with traditional taxonomic methods?

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CHAPTER OVERVIEW

8: Lab 6 - Ecosystems and Climate

[8.1: Overview and Purpose](#)

[8.2: Introduction to a Nature Walk](#)

[8.3: Part 1- Nature Walk Observations instructions](#)

[8.4: Part 2- Investigating the Effects of Climate Change on Local Ecosystems](#)

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8.1: Overview and Purpose

Overview and Purpose

This lab involves a nature walk to provide students an opportunity to spend time in nature and learn about the environment. The walk promotes local environmental awareness and encourages individuals to appreciate the natural world in terms of ecosystems, the interdependence of living organisms, climatic changes, and the importance of biodiversity.

Learning Outcomes

1. Increased knowledge and understanding of the local flora and fauna, including their characteristics, habitats, and ecological roles.
2. Improved awareness of the natural environment and the importance of preserving it for future generations.
3. Enhanced appreciation for the beauty and diversity of nature, leading to a more positive attitude towards the natural world.



Learning Objectives

- To provide participants with an opportunity to observe and learn about the local flora and fauna, including their characteristics, habitats, microclimates, and ecological roles.
- To promote environmental awareness and encourage participants to appreciate the natural world and its importance for human well-being.

Vocabulary

Producer

Primary consumer

Secondary consumer

Decomposer

Autotroph

Heterotroph

Food web

Photosynthesis

Cell respiration

Population

Community

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8.2: Introduction to a Nature Walk

Nature walks offer a unique opportunity to immerse yourself in the beauty and interconnectedness of the natural world and is a chance to make observations about living things that are easily missed or seemingly mundane. Whether you are a student, a scientist, or just someone who enjoys spending time outdoors, nature walks provide a chance to observe, learn, and reflect on the complex and interconnected relationships that exist within the ecosystem.

During a nature walk, you will have the chance to observe the plants, animals, and other living organisms that make up the environment. You will be able to see first-hand how they interact with each other and their surroundings and learn about the various adaptations that have allowed them to thrive in their specific environment.

By paying close attention to the details of your surroundings, you will develop a deeper appreciation for the complexity and interdependence of the natural world. You will also have the opportunity to make meaningful observations that will contribute to our broader understanding of the environment and the role that each component plays in it.

Please read the entire lab before going on your walk. Especially be sure to read **Part 1 and Part 2** because they give different instructions on things to look for on the walk. You will need to develop a data collection sheet in order to record your observations. By reading the lab and then creating a datasheet, you will have a great idea about the specific type of information you should be on the lookout for while walking.

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8.3: Part 1- Nature Walk Observations instructions

Going on a nature walk is a great opportunity to apply your knowledge and skills in a real-world setting.

Look at a map of your campus to identify the trail to be taken and identify potential areas of interest before you set foot outside.

OR

Choose a location: Pick a park, nature trail, or other natural area that you want to explore. Make sure the location is safe and suitable for the type of observations you want to make.

1. Gather your supplies: You will need a notebook, pen or pencil, thermometer, hand lens (or other magnifying device), and a camera or smartphone to take pictures of what you observe. You can also bring a field guide or app (such as Seek or Google Lens) to help identify plants and animals.
2. Start your walk: As you walk, look around and pay attention to the various plants, fungi, and animals in the area. Make note of the different types of vegetation and their growth patterns, as well as the different species of animals and their behaviors.
3. Make observations: Record your observations in your notebook, including descriptions of the physical characteristics of the plants and animals, as well as their behaviors and interactions with each other. Also, take pictures of anything interesting that you want to remember. You can insert these pictures into your notebook later.
4. Identify what you observe: Use a field guide or app to identify the plants and animals you see. Write down the names of the species you observe in your notebook, along with any other relevant information such as their classification and habitat.
5. Look for patterns: Look for patterns in your observations, such as the relationship between the types of vegetation and the species of animals present. Consider how the environment affects the habitat, behavior, and interactions of plants and animals.
6. Make hypotheses: Based on your observations, formulate hypotheses about the relationships between the different components of the ecosystem and how they may be affected by environmental factors such as temperature, rainfall, and light.
7. Analyze your data: After your nature walk, analyze your data and draw conclusions based on your observations and hypotheses. Consider how your findings relate to the larger scientific community's understanding of the ecosystem and its components.
8. Present your findings (optional): Present your findings to your classmates or submit a report to your teacher. Discuss the implications of your observations for the broader understanding of the ecosystem and the environment.

Activities for after the walk

Draw a food web including organisms that you saw in the same area, interacting together. Make sure to include a **producer**, at least **two levels of consumers**, and a **decomposer**. Label which organisms are **autotrophs** and which are **heterotrophs**. Use arrows to show how energy is transferred and lost throughout this web.

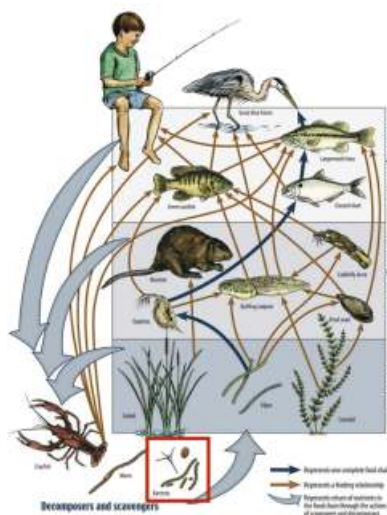


Figure 1. Example food web (CCBY 4.0 Kestin Schulz, Mariya W. Smit, Lydie Herfort and Holly M. Simon. The image (amended) was provided courtesy of the Missouri Department of Conservation.)

Food web drawing. Try to draw this somewhat as you saw it. For example, trees rooted in soil, bug on a leaf that is attached to grass, fungus growing from the ground, etc. Draw black arrows from the thing that might be eaten to the thing doing the eating.

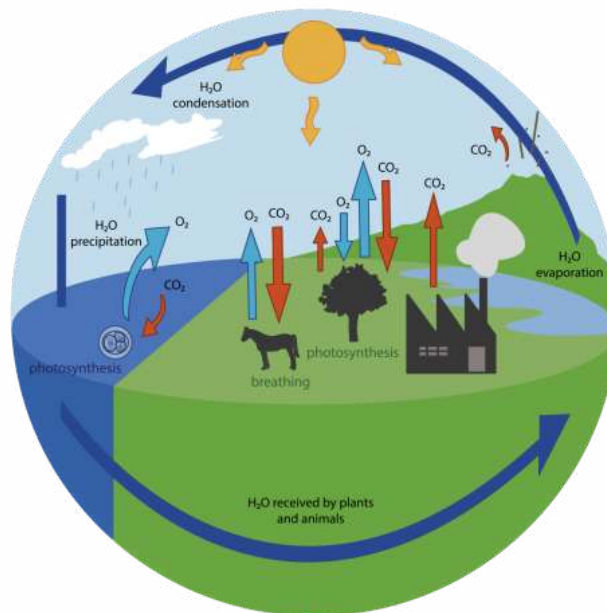
What is the ultimate source of energy for your food web?

What happens to this energy as it moves through the food web? Where does it end up?

In drawing your food web, you depicted a combination of the flow of energy and the cycling of nutrients through an ecosystem. Whereas energy has an input and an output, nutrients are continually recycled. These elements often cycle between the biotic and abiotic components in an ecosystem.

For example, water is composed of hydrogen and oxygen atoms. Water molecules precipitate from the atmosphere and fall as rain, saturating the soil. A plant absorbs those molecules through its roots, transporting them up to its leaves. Some of these water molecules will be broken apart during the process of photosynthesis, the oxygen exiting the plant as O_2 .

O_2 is gas that you will breathe in, while the hydrogen removed from the water might be used to form molecules of glucose that you will eat. The water not being broken apart exits the plant through its leaves, evaporating back into the atmosphere in a process called **transpiration**. The oxygen atoms that you breathed in will be used to release energy from your food (**cellular respiration**) and be joined back to hydrogen molecules to once again form water. This water might exit your body as vapor on your breath or as perspiration and return directly to the atmosphere. It might also soak into your clothes as sweat or exit as urine, be processed in a wastewater facility, then sent out to the ocean where it will evaporate back into the atmosphere.



Oxygen and carbon cycles through different processes. (CCBY 4.0 Wikimedia Commons as a contribution from an Art&Design School thanks of a collaboration between **Llotja** and **Amical Wikimedia**.)

The atoms of hydrogen and oxygen are broken apart and reassembled into other molecules multiple times in this process, but the overall outcome is a cycle, with atoms traveling between the **atmosphere, hydrosphere, lithosphere, and biosphere**, but rarely ever exiting our Earth system.

Using a blue pen or pencil, diagram the flow of water through the ecosystem around you in the food web you drew.

What is the role of plants in the global cycling of water?

Each species in an ecosystem is considered a **population**. This represents a group of actually or potentially interbreeding organisms and is an important area of study for both ecologists and geneticists. When multiple species in an ecosystem are considered together, this is called a **community**. For example, a forest could be composed of a single population of maple trees, but it is more likely to be a community, including other tree species like oak and pine, and animals like the whitetail deer or rainbow trout. All living organisms are considered to be a part of the community in a given ecosystem.

What is the difference between a community and an ecosystem?

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8.4: Part 2- Investigating the Effects of Climate Change on Local Ecosystems

Climate change is a global phenomenon that is affecting the environment and living organisms in various ways. The purpose of this part of the lab is to investigate the effects of climate on local ecosystems and understand the impact of potential human activities on the environment.

Materials:

- Field guide to local plants and animals
- Measuring tape or ruler
- Notebook and pen
- Camera or smartphone for taking pictures
- Weather app
- Thermometer

Procedure:

1. Follow the nature walk instructions from Part 1
2. Measure the normal outdoor air temperature and record it in your notebook along with observations about how you took this measurement such as 'shaded area' or 'strong wind'.
3. Take pictures of the environment to document different micro-ecosystems.
4. Measure the temperature of other habitats you observe including things like soil temperature, the temperature under a small shrub, the temperature of stream water, etc.
5. Analyze the data and make observations about changes in the temperature and the presence or absence of plants and animals between the different ecosystems you observed.
6. Draw conclusions about the effects that climate change may have on the ecosystem and discuss the impact of human activities on the environment.

Safety Precautions:

- Always follow safe outdoor practices, such as wearing appropriate clothing and footwear, staying on marked trails, and respecting wildlife.
- Avoid touching or disturbing plants and animals, as this can impact the ecosystem and disrupt the data collected.

Field Data Collection Pages

Insert more data pages as needed.

Attributions

Part 1: Maria Morrow (College of the Redwoods) CCBY 4.0 Merlot

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CHAPTER OVERVIEW

9: Lab 7 - Mammalian Anatomy Lab

[9.1: Overview and Purpose](#)

[9.2: Background Information on Mammals](#)

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9.1: Overview and Purpose

Overview and Purpose

For this lab students will create a 3D model of a fetal pig in order to study mammalian anatomy and compare it to human anatomy.

Learning Outcomes

1. Describe the anatomy of the pig as an example of a vertebrate mammal
2. Identify structures on the pig and know their functions
3. Identify structures that are a part of the digestive system, respiratory system, circulatory system, reproductive system, and urinary system
4. Compare and contrast pig anatomy to human anatomy



Learning Objectives

- Review general mammalian characteristics, body plans, and organ systems
- Construct a 3D model of the fetal pig

Vocabulary

Mammal

Monotremes:

Marsupials

Eutherians

Endothermic

Convergent evolution

Tetrapod

Vertebrates

Body plans

Body Cavity

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9.2: Background Information on Mammals

Mammalian Characteristics

Mammals, comprising about 5,200 species, are vertebrates that possess hair and mammary glands. Several other characteristics are distinctive to mammals, including certain features of the jaw, skeleton, integument, and internal anatomy. Modern mammals belong to three clades: monotremes, marsupials, and eutherians (or placental mammals).

The presence of hair, composed of the protein keratin, is one of the most obvious characteristics of mammals. Although it is not very extensive or obvious on some species (such as whales), hair has many important functions for most mammals. Mammals are endothermic, and hair traps a boundary layer of air close to the body, retaining heat generated by metabolic activity. Along with insulation, hair can serve as a sensory mechanism, which is particularly useful to nocturnal or burrowing mammals. Hair can also provide protective coloration or be part of social signaling, such as when an animal's hair stands "on end" to warn enemies, or possibly to make the mammal "look bigger" to predators.

Mammals, like birds, possess a four-chambered heart; however, the hearts of birds and mammals are an example of convergent evolution, since mammals clearly arose independently from different groups of tetrapod ancestors. Mammals also have a specialized group of cardiac cells (fibers) located in the walls of their right atrium called the sinoatrial node, or pacemaker, which determines the rate at which the heart beats. Mammalian erythrocytes (red blood cells) do not have nuclei, whereas the erythrocytes of other vertebrates are nucleated.

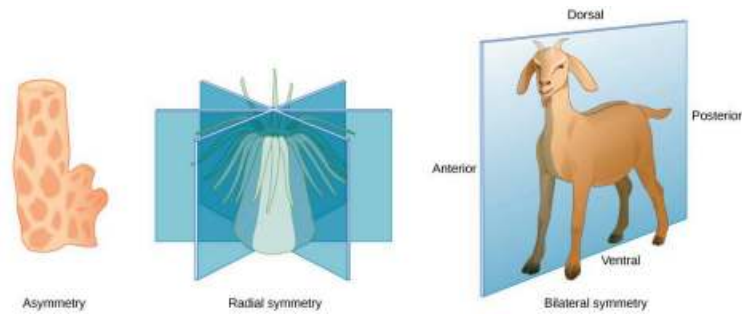
The kidneys of mammals have a portion of the nephron called the loop of Henle or nephritic loop, which allows mammals to produce urine with a high concentration of solutes—higher than that of the blood. Mammals lack a renal portal system, which is a system of veins that moves blood from the hind or lower limbs and region of the tail to the kidneys. Renal portal systems are present in all other vertebrates except jawless fishes. A urinary bladder is present in all mammals.

Mammalian brains also have certain characteristics that differ from the brains of other vertebrates. In some, but not all mammals, the cerebral cortex, the outermost part of the cerebrum, is highly convoluted and folded, allowing for a greater surface area than is possible with a smooth cortex. The optic lobes, located in the midbrain, are divided into two parts in mammals, while other vertebrates possess a single, undivided lobe. Eutherian mammals also possess a specialized structure, the corpus callosum, which links the two cerebral hemispheres together. The corpus callosum functions to integrate motor, sensory, and cognitive functions between the left and right cerebral cortexes.

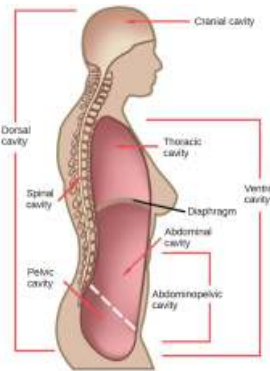
Eutherians (placentals) are the most widespread and numerous of the mammals, occurring throughout the world. Eutherian mammals are sometimes called "placental mammals" because all species possess a complex chorioallantoic placenta that connects a fetus to the mother, allowing for gas, fluid, and nutrient exchange. There are about 4,000 species of placental mammals in 18 to 20 orders with various adaptations for burrowing, flying, swimming, hunting, running, and climbing.

Body Plans

Animal body plans follow set patterns related to symmetry. They are asymmetrical, radial, or bilateral in form as illustrated below. Asymmetrical animals are animals with no pattern or symmetry; an example of an asymmetrical animal is a sponge. Radial symmetry describes when an animal has an up-and-down orientation: any plane cut along its longitudinal axis through the organism produces equal halves, but not a definite right or left side. This plan is found mostly in aquatic animals, especially organisms that attach themselves to a base, like a rock or a boat, and extract their food from the surrounding water as it flows around the organism. Bilateral symmetry is illustrated in the same figure by a goat. The goat also has an upper and lower component to it, but a plane cut from front to back separates the animal into definite right and left sides. Additional terms used when describing positions in the body are anterior (front), posterior (rear), dorsal (toward the back), and ventral (toward the stomach). Bilateral symmetry is found in both land-based and aquatic animals; it enables a high level of mobility.



Vertebrate animals have a number of defined body cavities. Two of these are major cavities that contain smaller cavities within them. The dorsal cavity contains the cranial and the vertebral (or spinal) cavities. The ventral cavity contains the thoracic cavity, which in turn contains the pleural cavity around the lungs and the pericardial cavity, which surrounds the heart. The ventral cavity also contains the abdominopelvic cavity, which can be separated into the abdominal and the pelvic cavities.



Within these body cavities you will find the organ systems. Organ systems are composed of various organs and structures with a common function. The chart below summarizes the major systems.

Organ Systems	Function
Digestive	Processing food involves ingestion (eating), digestion (mechanical and enzymatic breakdown of large molecules), absorption (cellular uptake of nutrients), and elimination (removal of undigested waste as feces).
Nervous	organism's control center: it processes sensory information from outside (and inside) the body and controls all behaviors—from eating to sleeping to finding a mate.
Endocrine	plays a role in growth, metabolism, and sexual development through the use of hormones
Musculoskeletal	provide support to the body and allow for a wide range of movement.
Respiratory	facilitate gas exchange by bringing in oxygen and getting rid of carbon dioxide
Circulatory	supplies the cells, tissues, and organs with oxygen and nutrients, and removes carbon dioxide and waste,
Urinary	function to filter blood and maintain the osmolarity of body fluids
Immune	diverse array of specialized cells and soluble molecules that coordinate a rapid and flexible defense system capable of providing protection against pathogens
Reproduction	Produce gametes and develops offspring

Fetal Pigs

Fetal pigs are commonly used as dissection specimens to represent mammalian anatomy. Nearly all the anatomical structures are similar between pig and human. Dissection is a powerful tool that provides us with a profound understanding of our own anatomy

and physiology as living, breathing creatures and also helps us to develop a stronger understanding of evolutionary relationships between taxonomic groups.

Materials needed:

1. Instructions provided by your professor
2. Color pencils
3. Scissors
4. Tape

Activity

For this lab we are going to create a 3D paper based model to study mammalian anatomy and compare it to human anatomy. Your professor will hand out the necessary materials and instructions for you to color and construct your own 3D pig model.

For Review Only, will not be directly linked to in final copy

[CLICK HERE!](#) To view the 3D paper based model activity

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Glossary

Allele | A variant form of a gene that is located at a specific position on a chromosome. Alleles can be dominant or recessive and can determine an organism's physical characteristics.

Analysis | The process of breaking down a problem into its component parts in order to understand it and solve it. Analysis involves identifying the problem, gathering data, and drawing conclusions.

Ancestral Trait | A trait that is shared between two species as a result of common ancestry. Ancestral traits are often used to determine the evolutionary relationships between species.

Autosomes | Chromosomes that are not involved in determining an organism's sex. Autosomes are numbered according to size, with the largest being chromosome 1 and the smallest being chromosome 22.

Autotroph | An organism that produces its own energy and nutrients through photosynthesis or chemosynthesis. Autotrophs are the primary producers in an ecosystem, and are the basis of all food webs.

Body Cavity | A space in an organism's body that contains organs and other internal structures. The most common body cavities are the coelom, the pseudocoelom, and the blastocoel.

Body Plans | The fundamental structure of an animal's body, including its symmetry, body segments, and appendages. Body plans are used to classify animals into different phyla.

Cell Respiration | The metabolic process by which cells break down food molecules to produce energy. Cell respiration involves the breakdown of glucose molecules to release energy in the form of ATP.

Cenozoic | The current geological time period, which began 66 million years ago and continues today. The Cenozoic is divided into three eras: the Paleogene, the Neogene, and the Quaternary.

Chromosome | Thread-like structures that consist of tightly condensed DNA (deoxyribonucleic acid), and associated proteins.

Clade | A group of species that share a common ancestor. Clades are used to classify organisms into different taxonomic groups.

Cladogram | A diagram that shows the evolutionary relationships between species. Cladograms are constructed using shared derived traits and can be used to determine the evolutionary history of a group of organisms.

Community | A group of organisms of different species that interact and share a common environment. Communities are composed of populations of different species and can be studied to understand the interactions between species.

Conclusion | A statement that summarizes the main points of an argument or experiment. Conclusions are based on the data collected and analyzed and should be supported by evidence.

Control | An experimental sample that serves as a reference for comparison with a test sample. Controls are used to ensure that the results of an experiment are not due to chance or other factors.

Controlled Variables | Factors that are held constant in an experiment to ensure that only the independent variable is changed. Controlled variables are essential for obtaining accurate results from an experiment.

Convergent Evolution | The process by which unrelated species evolve similar traits in response to similar environmental pressures. Convergent evolution is an example of natural selection and can result in similar adaptations in different species.

CRISPR/Cas9 | A cellular system within bacteria that acts as a defense mechanism to recognize and cleave invading DNA from bacterial viruses. This bacterial system is now used as technology that enables the efficient introduction of mutations into genes and organisms.

Data | Information that has been collected and analyzed. Data can be qualitative or quantitative and can be used to draw conclusions and make predictions.

Decomposer | An organism that breaks down dead or decaying organisms and returns their nutrients to the environment. Decomposers are essential for the cycling of nutrients in an ecosystem.

Dependent Variable | The factor that is measured in an experiment and is affected by the independent variable. The dependent variable is used to determine the effect of the independent variable on the system being studied.

Dichotomous Key | A tool used to identify unknown organisms by asking a series of questions with two possible answers. Dichotomous keys are used in taxonomy to classify organisms into different taxonomic groups.

Dominant allele | An allele that expresses the phenotype displayed in a heterozygous individual. The dominant allele masks the recessive allele.

DTC Testing | Direct-to-consumer genetic testing, which is used to assess an individual's risk of developing certain diseases. DTC testing can be used to identify genetic variations that may increase the risk of a disease or disorder.

Endothermic | An organism that produces its own heat. Endothermic organisms are able to maintain a constant body temperature and are able to survive in a wide range of temperatures.

Environmental Law | Laws that protect the environment from pollution, destruction, and other human activities. Environmental laws are designed to ensure that the environment is protected from harm.

Eutherians | A group of mammals that includes humans, cats, and dogs. Eutherians are distinguished from other mammals by their placenta and their ability to give birth to live young.

Experiment | A scientific procedure that is conducted to test a hypothesis. Experiments involve manipulating the independent variable and measuring the effect on the dependent variable.

Food web | A network of interconnected food chains in an ecosystem. Food webs are composed of producers, consumers, and decomposers, and can be used to understand the flow of energy in an ecosystem.

Fossil | The preserved remains or traces of ancient organisms. Fossils are used to study the evolution of life on Earth and can provide information about the environment in which the organism lived.

Gel Electrophoresis | A laboratory technique used to separate molecules based on their size and charge. Gel electrophoresis is used to separate DNA fragments, and can be used to identify genetic variations.

Gene | A unit of DNA that contains the instructions for producing a particular protein. Genes are passed from parent to offspring and can determine an organism's physical characteristics.

Gene therapy | The transfer of recombinant DNA into an organism to cause expression of a protein to treat a disease.

Genetic Disease | A disorder caused by a mutation in a gene. Genetic diseases can be inherited or acquired and can range from mild to severe.

Genetic Testing | Testing of a person's DNA to identify genetic variations that may cause or increase the risk of a disease or disorder. Genetic testing can be used to diagnose genetic diseases and to determine an individual's risk of developing certain diseases.

Genetically Modified Organism | An organism that receives recombinant DNA or whose genetic material has been altered, usually to cause expression of a protein from another organism.

Genome | The entire set of genetic information contained in an organism. The genome contains all of the genetic information necessary for an organism to function.

Genotype | The genetic makeup of an organism. The genotype is determined by the combination of alleles at each gene locus.

Heterotroph | An organism that must obtain its energy and nutrients from other organisms. Heterotrophs are the consumers in an ecosystem and include animals, fungi, and some bacteria.

Heterozygous | Having two different alleles at a particular gene locus. Heterozygous individuals have one dominant and one recessive allele, and the dominant allele is expressed in the phenotype.

Homologous chromosomes | In diploid organisms this is a pair of chromosomes that are the same length and contain the same genes in the same order on the chromosomal arms. One chromosome from the homologous pair was inherited from the mother and one chromosome from the homologous pair was inherited from the father.

Homozygous | Having two identical alleles at a particular gene locus. Homozygous individuals have two copies of the same allele, and the phenotype is determined by the allele.

Hydrophilic | Molecules with ionic or polar charges that dissolve easily in water.

Hypothesis | A proposed explanation for a phenomenon that can be tested through experimentation. Hypotheses are based on observations and can be tested using the scientific method.

Independent Variable | The factor that is changed in an experiment in order to observe the effect on the dependent variable. The independent variable is the variable that is manipulated in an experiment.

Linnaean System | A system of classification devised by Carl Linnaeus in the 18th century that organizes organisms into hierarchical categories based on their shared traits. The Linnaean system is still used today to classify organisms.

Mammal | A group of animals that are warm-blooded, have hair or fur, and feed their young milk. Mammals are divided into three groups: monotremes, marsupials, and eutherians.

Marsupials | A group of mammals that carry their young in a pouch. Marsupials are found mainly in Australia and South America and include kangaroos, koalas, and opossums.

Mesozoic | The geological period between 66 and 252 million years ago. The Mesozoic is divided into three periods: the Triassic, the Jurassic, and the Cretaceous.

Microarray | A laboratory tool that allows the detection of gene expression of thousands of genes simultaneously. Microarray slides (gene chips) contain DNA sequences that act as probes that will bind to complementary nucleotide sequences if the gene is expressed in the cells.

Monotremes | A group of mammals that lay eggs. Monotremes are found only in Australia and New Guinea and include the platypus and the echidna.

Mummification | The process of preserving a dead organism by exposing it to extreme drying. Mummification was practiced by the ancient Egyptians and was used to preserve the bodies of the deceased for the afterlife.

Paleozoic | The geological period between 541 and 252 million years ago. The Paleozoic is divided into six periods: the Cambrian, the Ordovician, the Silurian, the Devonian, the Carboniferous, and the Permian.

Phenotype | The physical and observable characteristics of an organism resulting from the interaction between its genetic makeup and the environment.

Pheromones | A chemical released by an individual of a species to communicate with or affect the behavior of other members of the same species.

Phylogenetic Tree | A diagram that illustrates the evolutionary relationships among a group of organisms, based on their similarities and differences in genetic, morphological, or behavioral traits.

Phylogeny | The evolutionary history and relationships among species or groups of organisms.

Pipette | A laboratory tool used to measure and dispense small volumes of liquid accurately.

Population | A group of individuals of the same species living in a particular geographic area and potentially interbreeding.

Primary consumer | An organism that feeds directly on producers in an ecosystem.

Producer | An organism that synthesizes organic compounds from simple substances and serves as the primary source of food for other organisms in an ecosystem.

Recessive Allele | An allele that is masked by the presence of a dominant allele and only expressed in the phenotype when paired with another recessive allele.

Recombinant DNA technology | Laboratory methods that enable one to join together DNA fragments from multiple organisms or multiple sources.

Scientific Method | A systematic and logical approach used by scientists to investigate phenomena, including making observations, formulating hypotheses, conducting experiments, and analyzing data.

Secondary consumer | An organism that feeds on primary consumers in an ecosystem.

Shared Derived Traits | Traits that are present in a group of organisms and inherited from a common ancestor but not found in more distantly related organisms.

Taxonomy | The branch of biology that deals with the identification, naming, and classification of organisms.

Tetrapod | A vertebrate animal with four limbs, or that descended from four-limbed ancestors.

Variables | Factors or conditions that can be changed or manipulated in an experiment.

Vertebrates | Animals with a backbone or spinal column, including fish, amphibians, reptiles, birds, and mammals.

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